The Faithful Machine: Trust, Delegation, and Algorithm Aversion in Financial Advice

Chris Tessone*

March 3, 2026

Working Paper — Do Not Circulate

Abstract

The algorithm aversion literature presents apparently contradictory findings: people sometimes reject algorithms that outperform humans, sometimes prefer them, and sometimes show no betrayal response to algorithmic failure. I argue these contradictions reflect a conceptual gap: the finance literature treats trust as a single dimension, conflating two structurally distinct modes. Drawing on philosophical trust theory (Baier, Holton, Hawley, Hardin), I distinguish *evaluative trust* (evidence-conditional performance assessment) from *delegative trust* (the grant of discretionary authority that is insensitive to ordinary performance signals but vulnerable to perceived betrayal). When delegative trust toward a category of trustee has never formed, I term this *delegative foreclosure*: a structural closure that no amount of performance evidence is likely to reopen. I test this framework with two studies. First, AI-assisted classification of 9,000 CFPB complaint narratives reveals a cross-product delegative gradient — mortgage complaints express significantly more delegative trust language than credit card complaints — while evaluative trust remains flat, consistent with dimensional separability. Second, a pre-registered experiment ($N = 354$) finds that delegative trust is the dominant predictor of willingness to use algorithmic advice ($\beta = -0.304$, $p < .001$), while evaluative trust is nonsignificant. These results suggest that the contradictions in the literature arise from an unmeasured dimension: findings attributed to algorithm aversion or appreciation operate within evaluative trust, while the delegative dimension that governs advisory relationships has gone unrecognized.

Keywords: trust, algorithm aversion, robo-advisors, delegative trust, evaluative trust, financial advice, behavioral finance, NLP, survey experiment

JEL Codes: G11, G41, G50, D83, D91

*Corresponding author. PhD student, University of Staffordshire. Email: t041965m@student.staffs.ac.uk. This paper was prepared for the UCLA Anderson Human × AI Finance Competition (Fink Center Conference on Financial Markets, April 2026). The author acknowledges extensive use of AI tools in research and writing; a detailed AI workflow description accompanies this submission.

Contents

1 Introduction **5**

2 Theoretical Framework: Evaluative and Delegative Trust **6**
2.1 The Distinction . . . 8
2.1.1 Evaluative Trust . . . 8
2.1.2 Delegative Trust . . . 8
2.1.3 Formal Characterization . . . 10
2.1.4 The Relationship Between the Two Modes . . . 11
2.2 The Agency Theory Bridge . . . 12
2.3 The "Peace of Mind" Reinterpretation . . . 13
2.4 Reconciling Algorithm Appreciation and Aversion . . . 14
2.5 The Transparency Paradox and Hypotheses . . . 15
2.5.1 Hypotheses . . . 16

3 Data and Methods **17**
3.1 CFPB Complaint Database Analysis . . . 17
3.1.1 Sample . . . 17
3.1.2 Classification Procedure . . . 18
3.1.3 Moral Foundations Validation . . . 18
3.1.4 Human Validation . . . 18
3.2 Prolific Experiment . . . 18
3.2.1 Design . . . 18
3.2.2 Participants . . . 18
3.2.3 Materials . . . 19
3.2.4 Measures . . . 19
3.2.5 Analytic Strategy . . . 20

4 CFPB Complaint Analysis: Results **21**
4.1 Sample and Classification Procedure . . . 21
4.2 Cross-Product Delegative Gradient . . . 21
4.3 Dual-Model Concordance . . . 22
4.4 Moral Foundations Validation . . . 23
4.5 Human Validation . . . 23
4.6 Dimensional Relationship . . . 24
4.7 Robustness: Narrative Length . . . 25
4.8 Exploratory: Company-Level Variation . . . 26
4.9 Summary . . . 26

5 Prolific Experiment: Results **26**
5.1 Sample and Exclusions . . . 26
5.2 Descriptive Statistics . . . 27
5.3 Experimental Manipulation . . . 28
5.4 Delegative Trust and Post-Shock Resilience . . . 29

5.5 Delegative Trust and Algorithm Aversion . . . 29
5.6 Financial Literacy Interaction . . . 31
5.7 Exploratory Analyses . . . 31
5.8 Summary . . . 32

6 Discussion **32**
6.1 What the Manipulation Failure Teaches . . . 32
6.2 Dimensional Separability Across Methods . . . 33
6.3 Reconciling Algorithm Aversion and Appreciation . . . 34
6.4 Delegative Foreclosure and Design Implications . . . 35
6.5 Limitations . . . 36

7 Conclusion **37**

A CFPB Classification Prompt **42**
A.1 System Prompt . . . 42
A.2 User Prompt Template . . . 43
A.3 Classification Parameters . . . 43

B Prolific Survey Instrument **44**
B.1 Part 1: Demographics and Financial Literacy . . . 44
B.1.1 Financial Literacy (Lusardi and Mitchell, 2014) . . . 44
B.1.2 Background Information . . . 45
B.2 Part 2: Condition Vignettes . . . 46
B.2.1 Condition A — Relationship/Fiduciary . . . 46
B.2.2 Condition B — Performance/Analytics . . . 46
B.2.3 Condition C — Neutral Control . . . 47
B.3 Part 3: Stage 1 Items (Evaluative and Delegative Trust) . . . 47
B.4 Part 4: Adverse Event . . . 47
B.5 Part 5: Comprehension Check . . . 47
B.6 Part 6: Stage 2 Items (Reactive Attitudes and Behavioral Response) . . . 48
B.7 Scale Reference . . . 48

C Inter-Rater Reliability Details **48**
C.1 Human Validation Sample . . . 49
C.2 Pairwise Categorical Agreement . . . 49
C.3 Pairwise Continuous Score Correlations . . . 49
C.4 Confusion Matrices . . . 50
C.5 Confidence-Stratified Agreement . . . 50
C.6 Classification Distribution Comparison . . . 51
C.7 Disagreement Pattern Analysis . . . 51
C.8 Illustrative Disagreement Cases . . . 51
C.9 Full-Sample Dual-Model Concordance ($N = 8{,}943$) . . . 52
C.10 Full-Sample Confidence-Stratified Concordance . . . 54

D Supplementary Tables **54**
D.1 CFPB Supplementary Tables 54
D.2 Prolific Supplementary Tables 57

1 Introduction

Why do some investors abandon algorithmic financial advisors after a single bad quarter while others remain patient through prolonged downturns? The behavioral finance literature offers two competing answers. Dietvorst et al. (2015) document "algorithm aversion" — the tendency to abandon algorithms after observing errors, even when the algorithm outperforms human alternatives. Logg et al. (2019) document the opposite: "algorithm appreciation," a systematic preference for algorithmic over human judgment in predictive domains. These findings have been treated as context-dependent results requiring reconciliation through moderator variables — task type, outcome observability, individual differences (for reviews, see Burton et al. 2020; Mahmud et al. 2022; Jussupow et al. 2024) — without a unifying theoretical account of why the same class of decision-aid produces opposite reactions.

We propose that the contradiction is an artifact of dimensional collapse. The finance literature treats trust as varying in degree but not in kind: investors trust more or less, along a single continuum. Drawing on philosophical accounts of trust that have not previously been applied to financial relationships — particularly the work of Baier (1986), Holton (1994), Nickel (2012), and Hawley (2014) — we distinguish two qualitatively different trust modes. *Evaluative trust* is evidence-conditional performance assessment: the investor monitors outcomes, compares them to benchmarks, and maintains the relationship insofar as the evidence supports it. Withdrawal takes the form of proportional disappointment. *Delegative trust* is the grant of discretionary authority with acceptance of opacity: the investor transfers judgment to the advisor and refrains from continuous verification. Violation produces not disappointment but betrayal — a reactive attitude whose intensity reflects the depth of the relationship rather than the magnitude of the performance shortfall.

This distinction resolves the algorithm aversion puzzle. Logg et al.'s appreciation and Dietvorst et al.'s aversion are both evaluative trust responding to performance signals — positive in one case, negative in the other. They are movements along the same dimension under different evidentiary conditions, not contradictory findings about human–algorithm interaction. Kormylo et al. (2026) introduce a third finding that completes the picture: betrayal aversion reduces delegation to human experts but not to algorithmic experts. Our framework explains why — betrayal is the reactive attitude specific to delegative trust violation, and under default experimental conditions, delegative trust toward algorithms does not form. The literature has been studying variation within evaluative trust without recognizing that a second dimension exists.

We test the evaluative/delegative framework using two complementary empirical approaches. First, we analyze 9,000 consumer complaint narratives from the Consumer Financial Protection Bureau (CFPB) database, classifying them for trust mode using dual-model AI classification (Claude 3.5 Sonnet and GPT-4o). The central finding is a cross-product delegative gradient: delegative trust indicators decline monotonically from mortgage to checking/savings to credit card complaints, tracking the theoretically predicted dimension of relationship opacity and duration, while evaluative trust indicators remain flat. This asymmetric pattern — one dimension varying while the other holds constant across product types — is the signature of dimensional separability.

Second, we conduct a pre-registered between-subjects experiment ($N = 354$ after pre-registered exclusions) in which participants encounter a robo-advisory platform under fiduciary, performance, or neutral framing before experiencing an identical adverse performance event. The manipulation does not shift trust modes — a theoretically informative failure we explain through Hardin's (1993) street-level epistemology

of trust, which holds that trust postures are learned through experience rather than installed by framing. However, the individual-differences analyses reveal the framework's predictive power: delegative trust is the dominant predictor of resistance to algorithm aversion ($B = -0.326$, $p < .001$) and resilience to exit ($B = -0.206$, $p = .001$), while evaluative trust predicts neither outcome.

The framework reframes what it means for an algorithmic advisor to be trustworthy. The "faithful machine" is not the one that maximizes benchmark-adjusted returns; it is the one whose institutional design permits the formation and honoring of delegative trust — the grant of discretionary authority that defines a fiduciary relationship.

Existing reviews have explicitly called for theory integration to resolve these contradictions (Burton et al., 2020; Jussupow et al., 2024); this paper answers that call. The paper makes three contributions. First, it introduces the evaluative/delegative distinction to behavioral finance, grounding it in philosophical trust theory and operationalizing it for empirical measurement. Second, it provides convergent evidence from naturalistic and experimental methods that the two trust modes are separable, respond to different signal classes, and predict distinct post-shock behaviors. Third, it resolves the algorithm aversion/appreciation contradiction by showing that both phenomena operate within the evaluative dimension, while the delegative dimension — absent from the experimental literature because standard paradigms do not create conditions for its formation — predicts which investors resist the aversion response. We introduce the concept of *delegative foreclosure* — the structural closure of a trust mode toward a category of trustee, derived from Hardin's insight about self-reinforcing distrust — to explain both the persistence of algorithm aversion in the face of superior performance evidence and the failure of institutional framing to shift trust postures.

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework, distinguishing evaluative and delegative trust, connecting them to the principal–agent literature and the Money Doctors model, and deriving five pre-registered hypotheses. Section 3 describes the data and methods for both empirical analyses. Section 4 reports the CFPB results. Section 5 reports the Prolific experimental results. Section 6 discusses implications, including the reconciliation of the algorithm aversion literature and the design implications of delegative foreclosure. Section 7 concludes.

2 Theoretical Framework: Evaluative and Delegative Trust

A foundational finding of modern financial economics is that trust matters for economic outcomes. Guiso et al. (2004, 2008) demonstrate that generalized trust predicts stock market participation, portfolio allocation, and the use of formal financial instruments, even after controlling for risk aversion, wealth, and education. The mechanisms linking trust to financial behavior have since received sustained empirical attention: Sapienza and Zingales (2012) distinguish between generalized social trust and specific beliefs about institutional trustworthiness; Gennaioli et al. (2015) model financial advisors as "money doctors" whose primary service is providing trust that reduces clients' perceived variance of returns; and Gurun et al. (2018) demonstrate that geographic and cultural proximity — channels plausibly related to trust — predict advisory relationships and their outcomes.

Despite the richness of these empirical contributions, the literature shares a common structural assumption: trust is treated as unidimensional. It varies in *degree* — investors trust more or less — but not in *kind*. The standard measurement approaches reflect this assumption. The World Values Survey item used by Guiso et al. asks whether "most people can be trusted" on a binary scale. The trust game paradigm introduced by Berg et al. (1995) measures trust as the dollar amount sent to an anonymous counterparty — a

continuous quantity representing the magnitude of risk accepted. Mayer et al.'s (1995) influential organizational framework decomposes trust into ability, benevolence, and integrity, but treats each as a continuous attribute of the trustee, not as a qualitatively distinct mode of the trust relationship itself.

These measurement approaches share a deeper assumption that Hardin (1993) identifies as a feature of "street-level" trust epistemology: initial trust is a learned Bayesian capacity that the trustor brings *to* the encounter, relatively independent of the specific trustee's properties. Before we accumulate evidence about a particular advisor or institution, we rely on priors learned from a category of interaction — prior experiences with financial professionals, narratives about algorithmic decision-making, cultural attitudes toward institutional authority. Hardin's insight reframes the measurement problem. Instruments that present a novel counterparty and measure the trustor's response are not capturing a reaction to that counterparty's characteristics; they are measuring the trustor's prior disposition toward the category. The question is not merely whether these instruments capture enough *variance* in trust — it is whether they capture the right *kind*. A trustor who approaches algorithms as tools to be assessed and a trustor who approaches them as agents to be entrusted with discretion are not expressing different magnitudes of the same attitude. They are operating in different modes, learned through different histories of encounter, and the standard instruments register both as a single quantity.

This unidimensional treatment creates an explanatory gap. Consider two welldocumented patterns in algorithmic finance. First, Dietvorst et al. (2015) find that individuals who observe an algorithm err — even when it outperforms human judgment on the same task — subsequently prefer human forecasters. This "algorithm aversion" appears to reflect a fragile form of trust in which a single negative signal triggers exit. Second, and apparently in tension, Logg et al. (2019) find that individuals systematically *prefer* algorithmic to human advice in predictive domains, exhibiting what they term "algorithm appreciation." A unidimensional model treats these as contradictory findings requiring reconciliation through moderator variables: perhaps aversion appears in some contexts and appreciation in others, depending on task type, outcome observability, or individual differences.

We propose that the contradiction is an artifact of dimensional collapse. Algorithm appreciation and algorithm aversion are not opposing ends of a single trust continuum; they are manifestations of different trust *modes* operating under different conditions. Investors who evaluate algorithmic performance against benchmarks are engaged in a qualitatively different trust relationship from investors who grant an algorithmic advisor discretionary authority over their portfolio. The former — which we term *evaluative trust* — is inherently conditional on continued satisfactory performance and produces rapid exit when performance falters. The latter — which we term *delegative trust* — involves a transfer of judgment that is not reducible to performance monitoring and produces a qualitatively different set of responses to adverse events, including both greater resilience under some conditions and a distinctive *betrayal* reaction under others.

The distinction between evaluative and delegative trust is not merely taxonomic. It generates predictions that the unidimensional model cannot. If trust varies only in degree, then higher-trust investors should uniformly exhibit greater resilience after adverse performance: they trusted more, so they tolerate more. But practitioners and emerging empirical evidence suggest a more complex pattern. Some investors with strong pre-shock trust relationships are among the *most* reactive to performance failures, displaying emotional responses — a sense of violated expectations, of misplaced faith — that exceed what a rational updating model would predict. The unidimensional framework has no vocabulary for this phenomenon. The evaluative/delegative framework does: these investors extended *delegative* trust, and its violation produces not the disappointment of a failed prediction but the betrayal of a breached relationship.

2.1 The Distinction

The evaluative/delegative framework draws on a substantial body of philosophical work on the nature of trust. Carter et al. (2020) demonstrated the relevance of philosophical trust concepts to financial contexts; the present paper provides the first empirical operationalization of these concepts with original data.

2.1.1 Evaluative Trust

Evaluative trust is the mode of trust that operates through assessment, verification, and conditional continuation. An investor operating in evaluative trust mode monitors portfolio performance, compares returns against relevant benchmarks, examines the rationale for specific allocation decisions, and maintains the advisory relationship conditional on continued evidence of competent management. The relationship is governed by evidence: when evidence supports the trustee's competence, the relationship continues; when evidence no longer supports it, the investor withdraws.

This mode of trust maps to well-established constructs in both philosophy and organizational behavior. Hardin (2002) models trust as "encapsulated interest": A trusts B because A believes that B's interests encapsulate A's interests. That is, B will perform well for A because doing so serves B's own goals. The trust is grounded in a prediction about behavior derived from an assessment of incentive alignment, and it is withdrawn when the incentive structure no longer supports the prediction. McAllister (1995) identifies a cognition-based dimension of interpersonal trust in organizations, grounded in evidence of the trustee's competence and reliability, which he distinguishes from an affect-based dimension grounded in emotional bonds and mutual care. Evaluative trust in financial contexts aligns closely with McAllister's cognition-based construct: it is rational, evidence-conditional, and does not require emotional attachment to the trustee.

Evaluative trust is not inherently shallow or deficient. Sophisticated evaluative trust — in which an investor develops nuanced frameworks for assessing risk-adjusted performance, understands factor exposures, and monitors for style drift — represents a high level of financial engagement. A client who retains an advisor because they have carefully evaluated the advisor's strategy and believe it sound is exercising evaluative trust. The key feature is not the depth of engagement but the *conditionality*: the relationship is maintained because and insofar as the evidence supports it.

The behavioral signatures of evaluative trust follow directly from its structure. Because evaluative trust is conditional on evidence, its withdrawal is proportional to the strength of disconfirming evidence and takes the form of *disappointment* — a cognitive assessment that expectations were not met. The investor's response to a performance shock is to seek information (Was the underperformance systematic or idiosyncratic? How did comparable strategies perform?), reassess the relationship, and either revise expectations or exit. The emotional register is analytical: the investor feels let down, not betrayed.

2.1.2 Delegative Trust

Delegative trust involves a qualitatively different relationship. An investor operating in delegative trust mode has granted the advisor — whether human or algorithmic — discretionary authority over decisions that the investor cannot or chooses not to fully evaluate. The investor has, in a meaningful sense, *transferred judgment*: not merely delegated execution of a strategy the investor has approved, but ceded the authority to determine what strategy is appropriate, to adjust it as conditions change, and to exercise professional judgment about matters the investor does not independently assess.

This mode of trust draws on several philosophical accounts that have not previously been applied to financial relationships. Baier (1986) argues that trust inherently involves accepted vulnerability: to trust is to be in a position where the trusted party can harm you, and to accept that position rather than eliminate it through control or monitoring. Nickel (2012), analyzing trust in medical contexts, characterizes it as entrusting someone with discretionary authority — authority to make decisions using their own judgment, not merely to execute pre-specified instructions. Hawley (2014) adds that trust involves a *commitment*: to trust someone is to rely on their commitment to do what they are relied upon to do, and to feel betrayal when that commitment is violated. The trusted party is not merely predicted to perform; they are counted on to fulfill an obligation.

Holton (1994) provides the critical behavioral distinction, drawing on Strawson's (1962) framework of reactive attitudes. He argues that what differentiates trust from mere reliance is the *reactive attitude* of the trustor upon violation. When I rely on my alarm clock and it fails, I am annoyed — but I do not feel betrayed, because the clock made no commitment and I invested no vulnerability. When I trust my financial advisor and she makes unauthorized trades, I feel betrayed — because I granted her discretionary authority, accepted the vulnerability that comes with not monitoring every decision, and she violated the commitment that this acceptance implied. The presence or absence of betrayal as a reactive attitude is, for Holton, the diagnostic marker of whether a relationship constitutes trust or mere reliance.

Applied to financial relationships, delegative trust manifests as acceptance of opacity (the investor does not insist on understanding every decision), tolerance of short-term uncertainty (the investor does not react to every quarterly fluctuation), and a distinctive *scope* of the relationship that extends beyond narrow performance metrics. The finding by Rossi and Utkus (2020) that advisory clients cite "peace of mind" and "delegation" as primary reasons for hiring advisors points toward delegative trust, though the existing literature does not isolate it from evaluative trust.

The behavioral signatures of delegative trust differ sharply from those of evaluative trust, particularly in response to adverse events. Because the investor has accepted vulnerability and granted discretionary authority, violation produces not proportional disappointment but *betrayal* — a response whose intensity reflects the depth of the trust relationship rather than the magnitude of the performance shortfall. The betrayal response is qualitatively distinct: the investor feels not merely that their predictions were wrong, but that their vulnerability was exploited, that the commitment they relied upon was hollow. This is the phenomenon that the unidimensional model cannot capture — the investor who was *most* trusting in the delegative sense may produce the *most* intense negative reaction, because they had the most to lose.

A further distinction concerns what happens *between* adverse events. An evaluative trustor monitors continuously; their trust is maintained through ongoing verification. A delegative trustor monitors less — not from inattention or ignorance, but from a *choice* to refrain from verification that would convert the relationship from delegation to surveillance. This distinction generates a testable prediction: platforms that encourage frequent performance monitoring may inadvertently shift users from delegative to evaluative trust mode, increasing the likelihood of exit after subsequent performance shocks. Pettit (1995) argues that introducing checks and surveillance removes the opportunity to manifest trusting reliance, thereby foreclosing the trust relationship itself; the monitoring that was meant to ensure compliance prevents trust from forming in the first place. The investor who checks their portfolio daily is, through the act of checking, operating in evaluative mode — regardless of how the relationship was initially framed.

Delegative trust also has a distinctive failure mode that is not simply the inverse of its formation. Hardin (1993) observes that learned extreme distrust is self-reinforcing: it forecloses the encounters that might up-

date the prior. A person who has developed generalized distrust of a category of agent avoids interactions with members of that category, ensuring they never accumulate the experience that could revise the disposition. Applied to our framework, this produces a phenomenon we term *delegative foreclosure*: the closure of an entire trust mode toward a category of trustee, not the downward revision of a trust estimate within that mode. A consumer who has had a negative experience with one algorithmic advisor — or who has absorbed sufficiently negative narratives about the category — does not merely lower their evaluative assessment of algorithmic advice. They foreclose the *possibility* of extending delegative trust to any algorithmic system. The evaluative channel remains open: they can still be shown performance data, benchmark comparisons, and track records, and they may revise their evaluative trust accordingly. But the delegative channel is closed. No amount of performance evidence is likely to reopen it, because performance evidence addresses the wrong dimension.

Delegative foreclosure is categorically different from algorithm aversion as the existing literature describes it. Dietvorst-style aversion is an evaluative phenomenon: the user observed poor performance and updated their competence assessment downward. Foreclosure is a structural change in the trustor's available modes of relating to algorithmic trustees. This distinction matters practically because the interventions appropriate to each are different. Evaluative aversion can be addressed through better performance data, transparency, and track records. Foreclosure requires a different kind of repair — the reconstruction of conditions under which the trustor can experience vulnerability as safe, which is a relational project, not an informational one.

2.1.3 Formal Characterization

The preceding accounts can be stated with greater precision. Let investor i's trust relationship with advisor j in financial domain D be characterized by an ordered pair (E, δ), where each component captures a distinct dimension of the relationship.

Definition 1 (Evaluative Trust). $E(i, j, D) \in [0, 1]$ is investor i's evidence-conditional assessment of advisor j's competence in domain D. Let s_t denote a performance signal received at time t (e.g., portfolio returns relative to a benchmark). Evaluative trust updates in response to performance information:

$$E_t = f(E_{t-1}, s_t), \quad \text{where } \frac{\partial f}{\partial s} > 0.$$

The function f is monotonically increasing in the signal: positive performance signals raise evaluative trust; negative signals lower it. When E_t falls below investor i's continuation threshold E^*, the investor exits. The reactive attitude upon exit is *disappointment*, proportional to the magnitude of the shortfall $(E^* - E_t)$.

Definition 2 (Delegative Trust). $\delta(i, j, D) \in [0, 1]$ is the degree to which investor i has granted advisor j discretionary authority over decisions in D that i does not independently verify. Unlike evaluative trust, delegative trust is not updated by performance signals alone:

$$\frac{\partial \delta}{\partial s_t} \approx 0 \quad \text{for ordinary performance signals.}$$

Delegative trust responds instead to *relational signals* — evidence about the commitment structure itself. Let r_t denote a relational signal (e.g., discovery of a fiduciary breach, unauthorized exercise of discretion,

institutional restructuring that alters the authority relationship). Then:

$$\delta_t = h(\delta_{t-1}, r_t), \quad \text{where } \frac{\partial h}{\partial r} \text{ is large and negative for commitment violations.}$$

The reactive attitude upon violation is *betrayal*, whose intensity reflects the prior level of delegative trust: a high-δ investor who experiences a relational violation produces a stronger negative response than a high-E investor who experiences an equivalent performance shock, because what has been violated is not a prediction but a grant of discretionary authority.

These definitions generate three claims that the subsequent empirical analyses test.

Claim 1 (Dimensional Separability). E and δ respond to different signal classes. Across contexts that vary the salience of performance information versus relational commitment, E and δ should vary independently. Formally: a manipulation that increases the salience of performance monitoring (e.g., benchmark framing) should raise E without proportionally raising δ, while a manipulation that emphasizes fiduciary commitment should raise δ without proportionally raising E. A unidimensional construct would require both to move together.

Claim 2 (Asymmetric Post-Shock Response). After an adverse event $s_t < \bar{s}$:

High-E investors: $\Pr(\text{exit}) = g(\bar{s} - s_t, E_{t-1})$ — exit probability increasing in shortfall magnitude.

High-δ investors: $\Pr(\text{exit})$ is low when s_t carries only performance information, because $\partial\delta/\partial s_t \approx 0$. But if s_t also conveys relational information r_t (the advisor failed to exercise genuine judgment, the institutional commitment was hollow), then $\Pr(\text{exit})$ can *exceed* the evaluative case, accompanied by betrayal rather than disappointment.

This generates what we term the betrayal premium prediction — related to the pattern documented by Kormylo et al. (2026), but reframed through the evaluative/delegative distinction: for investors in the delegative mode, the gap between betrayal intensity and disappointment intensity should be positive after an adverse event, while for evaluative investors it should be near zero or negative.

Claim 3 (The Transparency Paradox). Let τ represent the degree of transparency or auditability in the advisory relationship. Increasing τ shifts the trust relationship along different dimensions:

$$\frac{\partial E}{\partial \tau} > 0 \quad \text{(transparency makes performance verifiable, supporting evaluative trust)}$$
$$\frac{\partial \delta}{\partial \tau} < 0 \quad \text{(salient, frequent monitoring erodes the discretionary space in which delegative trust operates)}$$

This is the mechanism underlying O'Neill's (2002) diagnosis that audit culture undermines trust, formalized through the two-dimensional framework. A unidimensional model predicts that transparency uniformly increases trust; the two-dimensional model predicts that it increases one component while decreasing the other — with net effects on investor behavior that depend on which mode dominates the relationship.

2.1.4 The Relationship Between the Two Modes

The formal characterization makes explicit that evaluative and delegative trust are distinct dimensions, not opposite poles of a single continuum. In principle, an investor can exhibit high levels of both: carefully evaluating an advisor's track record (high E) while simultaneously granting discretionary authority over

tactical decisions within an approved strategic framework (high δ). An investor can also exhibit low levels of both — maintaining a financial relationship through inertia without either evaluating performance or granting meaningful discretionary authority.

However, the two modes are not fully independent in practice. In constrained communicative contexts — including the complaint narratives analyzed in this study — the two modes compete for expression. A narrative dominated by relational language (betrayal, violated expectations, exploited vulnerability) leaves less space for analytical language (benchmark comparison, performance metrics, cost assessment), and vice versa. This produces an inverse correlation in observed expression that reflects the communicative constraint rather than a structural opposition between the constructs themselves. The critical test of dimensional separability is not the within-context correlation but the *cross-context asymmetry* predicted by Claim 1: whether the two dimensions vary independently across different conditions. As we demonstrate in the empirical analysis, delegative trust expression varies substantially across financial product categories while evaluative trust expression remains essentially constant — a pattern that a single bipolar dimension cannot produce.

The distinction between evaluative and delegative trust also clarifies a long-standing measurement problem. McAllister's (1995) affect-based trust items — developed for interpersonal workplace relationships — fail when applied to algorithmic trustees because they operationalize the relational dimension of trust through interpersonal warmth ("We have a sharing relationship"; "I can talk freely to this individual about difficulties"). This failure has been interpreted as evidence that algorithms cannot receive trust beyond the competence-based dimension.[1] The most comprehensive review of trust in AI (Glikson and Woolley, 2020) similarly distinguishes cognitive from emotional trust, but treats these as antecedent categories rather than as structurally independent modes with distinct reactive attitudes and update dynamics. We argue that the conclusion is premature. The failure is one of *operationalization*, not of *construct*: delegative trust with algorithmic systems operates not through affective bonds but through institutional features — fiduciary framing, regulatory protections, discretionary authority structures — that create the conditions for vulnerability acceptance and judgment transfer without requiring interpersonal warmth. In terms of the formal characterization, institutional features can raise δ even when affective channels are absent, provided they create credible commitment structures. The Prolific experiment reported in this paper tests this claim directly.

2.2 The Agency Theory Bridge

The evaluative/delegative distinction maps onto a foundational tension in financial economics: the spectrum between complete and incomplete contracting. Under Jensen and Meckling's (1976) framework, the principal–agent relationship is governed by a contract that specifies observable performance metrics, monitoring mechanisms, and enforcement provisions. The principal's problem is one of verification — ensuring the agent acts as contracted — and the solution is better information. This is evaluative trust rendered in the language of corporate finance: the investor trusts the advisor because and insofar as performance is observable, contractible, and enforceable.

But real advisory relationships are not complete contracts. Grossman and Hart (1986) demonstrate that when contingencies cannot be fully specified in advance — when the space of possible market states exceeds

[1] See, e.g., the discussion in Lee and See (2004) and Muir (1994) on the limits of trust in automated systems. Madhavan and Wiegmann (2007) explicitly contrast human–human trust (which includes relational dimensions) with human–automation trust (which they characterize as primarily performance-based).

the contracting parties' ability to enumerate them — residual control rights must be allocated. Someone must decide what to do in circumstances the contract does not address. Delegative trust, in this framing, is the investor's acceptance of residual control rights held by the advisor: the authority to exercise judgment over decisions that were not and could not have been contracted upon. The investor does not merely delegate execution of a pre-specified strategy; they cede the right to determine what strategy is appropriate as conditions change. The trust inheres not in the ability to verify outcomes but in the willingness to grant discretion over uncontracted states of the world.

This mapping carries a methodological implication. The trust game introduced by Berg et al. (1995) — the dominant experimental paradigm for measuring trust in economics — is structurally a complete-contracting environment. The payoff matrix is known, the interaction is typically one-shot, the counterparty is anonymous, and the decision reduces to a single monetary transfer with a calculable expected return. Multi-round and repeated-interaction variants exist (Johnson and Mislin, 2011), but even these retain the core feature of observable, calculable payoffs — the trustor always knows the outcome of each round and can condition future behavior on it. These are precisely the conditions under which evaluative trust operates: the trustor assesses the probability of reciprocation, calibrates the amount sent accordingly, and withdraws from the interaction once the outcome is observed. There is no residual control, no unspecified contingency, no discretionary authority to grant. If the finance literature validates its trust measures against trust game behavior — and Johnson and Mislin's (2011) meta-analysis documents extensive use of the paradigm across disciplines — then the empirical foundation may be systematically measuring evaluative trust while the advisory relationships it seeks to explain are substantially delegative.

2.3 The "Peace of Mind" Reinterpretation

The evaluative/delegative framework's explanatory power is most clearly demonstrated against the most influential model of trust in financial advice. Gennaioli et al. (2015) propose that financial advisors function as "money doctors" whose primary service is the provision of trust. In their model, trust reduces the client's perceived riskiness of the investment, operating through effective risk aversion rather than beliefs about return distributions: the advisor does not necessarily improve expected outcomes or alter the client's perception of variance, but the client experiences the portfolio as less risky because a trusted professional is managing it. Clients pay advisory fees not for alpha but for "peace of mind." The model is elegant, empirically productive, and — we argue — ambiguous on a dimension that determines its behavioral predictions.

The ambiguity is this: what kind of trust reduces perceived variance? Under an evaluative interpretation, the advisor provides peace of mind through optimization and legibility. The client monitors less frequently not because they have transferred judgment but because the advisor's demonstrated competence has lowered the expected incidence of events worth monitoring. The client's comfort is evidence-conditional: it persists because performance has been satisfactory and would dissolve if performance deteriorated. Under a delegative interpretation, peace of mind arises from the act of judgment transfer itself. The client experiences reduced variance not because outcomes are objectively smoother but because the burden of decision-making — of choosing, second-guessing, and bearing responsibility for allocation choices — has been lifted. The comfort comes from the relationship, not from verification of the relationship's outputs.

These interpretations generate sharply different predictions about what happens after a performance shock. The evaluative peace-of-mind client experiences the shock as a variance event that exceeds prior

expectations. Their response is to reassess: Was the advisor's strategy flawed? Did the underperformance reflect systematic failure or market conditions? The emotional register is disappointment calibrated to the magnitude of the shortfall relative to expectations. If the evidence warrants it, they exit; if it does not, they revise expectations and continue. The delegative peace-of-mind client experiences the shock differently. If the advisory relationship remains intact — if the advisor communicates, explains the approach going forward, and continues to exercise the discretionary authority the client granted — the client's response may be patience. The advisor is navigating difficulty on the client's behalf, which is precisely what they were trusted to do. But if the shock reveals that the delegative relationship was hollow — that the advisor was not exercising genuine judgment, or that the institutional structure does not support the discretionary authority the client believed they had granted — the response is betrayal, not disappointment. The intensity of the reaction reflects the depth of the prior vulnerability, not the magnitude of the financial loss.

Survey evidence suggests the delegative interpretation better describes advisory clients' self-reported experience. Rossi and Utkus (2020) find that clients cite "delegation" and "peace of mind" as primary reasons for retaining advisors — language that maps more naturally to judgment transfer than to variance assessment. Consistent with this view, Foerster et al. (2017) find that advisors' own portfolios exhibit the same biases they recommend to clients, suggesting that what advisors provide is not superior information but rather a relationship that enables clients to bear risk they would otherwise avoid. But the existing literature does not isolate the two mechanisms. The Money Doctors model is agnostic about the *kind* of trust that does the variance-reducing work, and its predictions are correspondingly silent about the qualitative differences in post-shock behavior that the evaluative/delegative distinction reveals. Our framework does not replace the Gennaioli et al. model; it disaggregates a parameter the model treats as unitary.

2.4 Reconciling Algorithm Appreciation and Aversion

Three findings in the algorithmic decision-making literature appear to be in tension. Logg et al. (2019) find that individuals systematically prefer algorithmic to human advice in predictive domains, exhibiting "algorithm appreciation." Dietvorst et al. (2015) find that individuals who observe an algorithm err subsequently refuse to use it, even when it outperforms human forecasters — "algorithm aversion." And Bigman and Gray (2018) find that individuals resist algorithmic involvement in decisions with moral dimensions, even when the algorithm's judgments are equivalent to or better than human ones. The literature has treated these as context-dependent findings requiring reconciliation through moderator variables (e.g., Castelo et al., 2019; Jussupow et al., 2024): appreciation prevails in some task environments, aversion in others, depending on outcome observability, task subjectivity, and the moral valence of the decision.

The evaluative/delegative framework offers a more parsimonious resolution. As documented in the existing literature, algorithm appreciation is evaluative trust operating as expected: when an algorithm demonstrates predictive competence in a domain where performance is observable and quantifiable, evaluative trust increases. The algorithm is assessed, verified, and preferred precisely because it outperforms. Algorithm aversion after errors is evaluative trust failing as expected: the same performance-conditional relationship that produced appreciation now produces withdrawal when a negative signal arrives. The trust was never resilient to error because it was never grounded in anything beyond continued evidence of competence. Both findings, as the literature describes them, document evaluative trust under different performance conditions, not opposing orientations toward algorithms. Our framework leaves open the possibility that delegative trust toward algorithms can also form — as the Prolific experiment tests directly — but the existing appreciation/aversion literature has not studied this dimension.

Resistance to algorithmic moral decisions is a different phenomenon entirely. Moral decisions require precisely the exercise of discretionary judgment that defines delegative trust: they cannot be reduced to optimization over a known objective function, they require weighing incommensurable values, and they implicate the decision-maker's character — their integrity, not merely their competence. Bigman and Gray's finding may reflect, at least in part, resistance to extending *delegative* trust to algorithms, rather than a generalized aversion. Respondents who willingly delegate predictive tasks to an algorithm (evaluative trust) balk at delegating moral judgment to the same system (delegative trust). The trust mode, not the trustee, has changed. Similarly, Longoni et al. (2019) find that resistance to medical AI is driven by "uniqueness neglect" — a concern that algorithms cannot account for one's particular circumstances — which in the present framework reflects a delegative concern: the desire for an advisor who exercises discretion about the client's specific situation, not merely one who processes inputs accurately.

The concept of delegative foreclosure introduced above explains a further puzzle in this literature: the persistence of algorithm aversion even when users are presented with compelling evidence of algorithmic superiority. If the resistance were purely evaluative — a low competence assessment — then strong performance data should resolve it. Dietvorst et al. (2015) find that it does not. Our framework identifies the mechanism: for users who have foreclosed delegative trust toward algorithms, performance evidence addresses the evaluative dimension while the operative barrier is structural — the delegative channel is closed, and no volume of evidence within the evaluative channel is likely to reopen it. The appropriate intervention is not more data but the reconstruction of conditions under which delegation feels safe.

Dietvorst et al.'s (2018) follow-up study provides further support. They find that even trivial modification ability — the capacity to adjust an algorithm's output by a small amount before it takes effect — substantially reduces algorithm aversion. This result is puzzling under a pure competence account: a minor adjustment does not meaningfully improve the algorithm's performance, so why should it change the user's willingness to rely on it? Under the evaluative/delegative framework, the answer is immediate. Modification grants the user a form of discretionary authority over the final output. In Nickel's (2012) terms, the user is no longer a passive recipient of algorithmic decisions but a participant who exercises judgment — however minimally — over the outcome. This converts the relationship from pure evaluation (observing and assessing the algorithm's performance) to partial delegation (sharing authority over the final decision). The trust changes in kind, not merely in degree, and the behavioral consequences follow: a user who has exercised even token authority over a decision is less likely to experience the algorithm's error as a failure imposed upon them and more likely to experience it as a joint outcome for which they bear partial responsibility.

2.5 The Transparency Paradox and Hypotheses

The evaluative/delegative framework generates a counterintuitive prediction about a central policy question in financial regulation: the role of transparency. O'Neill (2002, 2014) argues that the contemporary culture of transparency — the proliferation of disclosure mandates, performance metrics, audit requirements, and accountability mechanisms — has failed to restore public trust in institutions despite being designed expressly for that purpose. Her diagnosis is that transparency and audit culture cannot solve a trust crisis, and may in fact deepen it. The mechanisms of accountability, intended to make institutions more trustworthy, have instead produced "a culture of suspicion" in which institutional actors optimize for measurability rather than the underlying goods they are meant to deliver.

O'Neill's diagnosis is acute but theoretically incomplete. She identifies the failure without fully specifying the mechanism by which transparency undermines trust. The evaluative/delegative framework supplies this mechanism. Transparency builds evaluative trust: it makes the trustee's actions observable, verifiable, and assessable against explicit criteria. Disclosure mandates, benchmark reporting, and performance analytics all serve to make the advisory relationship more legible to the investor. But transparency simultaneously erodes the conditions under which delegative trust operates. Delegative trust requires discretionary space — the authority to exercise judgment over matters that the principal has chosen not to fully monitor. When every decision must be disclosed, justified, and subjected to audit, the discretionary space collapses. The advisor no longer holds residual control rights; they execute under surveillance. And an investor who verifies every decision is, by definition, operating in evaluative mode — regardless of how the relationship was initially framed or intended.

Applied to algorithmic finance, this analysis yields a specific prediction: explainability mandates for robo-advisors — regulatory requirements that algorithms disclose the reasoning behind their investment decisions — may paradoxically *increase* algorithm aversion rather than reduce it. By making algorithmic decisions transparent and auditable, regulators shift users from a delegative trust equilibrium (in which they accept the algorithm's discretionary authority and tolerate short-term uncertainty) to an evaluative trust equilibrium (in which they assess each decision against their own expectations and benchmarks). The shift increases the user's sensitivity to short-term performance fluctuations and lowers the threshold for exit — precisely the opposite of the regulatory intent. Mandatory performance disclosure may induce monitoring behavior that converts the long-term commitment benefits documented by Gennaioli et al. (2015) into the short-term reactivity documented by Dietvorst et al. (2015).

Carter et al. (2020) identified the need for philosophical trust theory in financial contexts, applying Baier's vulnerability account and Hawley's commitment framework to the 2008 banking crisis. Their analysis demonstrated that reactive attitudes — betrayal, resentment, a sense of violated expectations — distinguish trust failures from mere reliance failures in financial relationships. But their contribution remained at the level of diagnosis: they showed that philosophical trust concepts apply to finance without operationalizing the distinction, without testing whether institutional framing shifts investors between trust modes, and without extending the framework to the algorithmic context that now dominates retail financial advice. The present paper answers all three.

2.5.1 Hypotheses

The theoretical framework developed across the preceding sections generates five pre-registered predictions, tested through a between-subjects experiment in which participants are randomly assigned to one of three conditions — fiduciary framing (Condition A), performance framing (Condition B), or a neutral control (Condition C) — before encountering an identical adverse performance event.

H1 (Trust Mode Separation). Fiduciary framing produces higher delegative trust than performance framing; performance framing produces higher evaluative trust than fiduciary framing. If evaluative and delegative trust are distinct dimensions responsive to different institutional features, then framing that emphasizes discretionary authority and relational commitment (Condition A) should elevate delegative trust, while framing that emphasizes verifiable performance and benchmark comparison (Condition B) should elevate evaluative trust.

H2 (Betrayal Premium). The difference between betrayal and disappointment responses after the adverse event is larger under fiduciary framing than under performance framing. Delegative trust violation

produces betrayal (Holton, 1994); evaluative trust failure produces disappointment. If Condition A successfully induces delegative trust, the betrayal response should be elevated relative to disappointment — a pattern we term the "betrayal premium."

H3 (Algorithm Aversion Mechanism). The tendency to switch from algorithmic to human advice after the adverse event is predicted by evaluative trust scores but not by delegative trust scores. Algorithm aversion, in the Dietvorst sense, is an evaluative trust phenomenon: performance-conditional trust withdrawn after a negative signal. Delegative trust, grounded in accepted vulnerability and granted discretion rather than in performance verification, should not predict the algorithm-to-human switch.

H4 (Delegative Resilience). Higher delegative trust scores predict lower exit-seeking behavior after the adverse event, controlling for evaluative trust. Delegative trust tolerates short-term uncertainty because the trustor has accepted the trustee's exercise of discretionary authority through adverse conditions — the advisor is navigating difficulty, which is what they were trusted to do.

H5 (Financial Literacy Interaction). The treatment effect of fiduciary versus performance framing on delegative trust is concentrated among high-literacy respondents. Low-literacy respondents default to elevated delegative scores across conditions — not from trust, but from the absence of evaluative capacity. High-literacy respondents *can* evaluate but, under fiduciary framing, are given institutional permission not to. The interaction distinguishes genuine delegation — a choice made by those with the capacity to do otherwise — from delegation by default.

3 Data and Methods

This study employs two complementary empirical approaches: a naturalistic analysis of consumer complaint narratives from the CFPB database and a pre-registered between-subjects experiment conducted on the Prolific platform. The two analyses are methodologically independent — different populations, product domains, measurement instruments, and analytic strategies — and are designed to provide convergent evidence for the evaluative/delegative trust framework through triangulation rather than mutual dependence.

3.1 CFPB Complaint Database Analysis

3.1.1 Sample

We sampled 9,000 consumer complaint narratives from the CFPB database, equally distributed across three product categories: mortgage (n = 3,000), checking and savings (n = 3,000), and credit card (n = 3,000). These categories were selected to vary along the dimension our framework identifies as theoretically relevant: the degree to which the consumer–institution relationship involves delegation of discretionary authority. Mortgage relationships are long-duration, high-stakes, and involve ongoing servicing decisions (escrow management, modification processing, loss mitigation) that consumers cannot easily monitor. Credit card relationships are largely transactional, with observable terms and assessable performance on standard metrics (interest rates, fees, dispute resolution). Checking and savings accounts fall between these poles. The sample was drawn from narratives with sufficient text for classification (minimum 50 words) filed between 2018 and 2024.

3.1.2 Classification Procedure

Each narrative was independently classified by two large language models — Claude 3.5 Sonnet (Anthropic; `claude-3-5-sonnet-20241022`) and GPT-4o (OpenAI) — using identical classification prompts. The prompts defined the evaluative and delegative trust constructs using language drawn from the theoretical framework and asked each model to provide three outputs for each narrative: (a) a categorical classification (evaluative, delegative, or unclassifiable), (b) continuous scores on each dimension (0–1 scale), and (c) a confidence rating (0–1 scale). The dual-model approach serves as an automated inter-rater reliability check: agreement between independently developed language models, trained on different data and using different architectures, provides stronger evidence for classification validity than either model alone.

Of the 9,000 narratives, Claude 3.5 Sonnet produced valid classifications for 8,991 (99.9%) and GPT-4o for 8,943 (99.4%). All dual-model analyses use the $N = 8{,}943$ narratives successfully classified by both models.

3.1.3 Moral Foundations Validation

As an independent construct validity check, all narratives were scored using the extended Moral Foundations Dictionary (eMFD; Hopp et al., 2021), which provides continuous scores on five moral foundations: care, fairness, loyalty, authority, and sanctity. The theoretical prediction is that delegative trust violations should activate relational moral foundations — particularly loyalty, the foundation most closely associated with betrayal and in-group allegiance — at higher rates than evaluative trust complaints.

3.1.4 Human Validation

I independently classified a confidence-stratified subsample of 100 narratives to assess whether AI classifications capture distinctions recognizable to a human coder. Both models reported high confidence overall: Sonnet's mean confidence was 0.882 and GPT-4o's was 0.877, with 68.8% of Sonnet classifications at confidence ≥ 0.90. To ensure the human validation sample included difficult cases, the subsample was stratified by Sonnet confidence level (30 low-confidence, 40 medium, 30 high-confidence) at relatively generous thresholds rather than sampling proportionally from the confidence distribution.

3.2 Prolific Experiment

3.2.1 Design

The experiment employed a between-subjects design with three conditions. Participants were randomly assigned to encounter a hypothetical robo-advisory platform (WealthPath) under one of three framings: fiduciary/relationship (Condition A), performance/analytics (Condition B), or neutral control (Condition C). All participants then experienced an identical adverse performance event before completing dependent measures. The study was pre-registered at AsPredicted (#275,531).

3.2.2 Participants

Demographics and the financial literacy battery were administered before condition assignment to prevent the vignette from priming financial literacy responses.

We recruited 471 participants through the Prolific platform, targeting U.S.-based adults with balanced demographic representation. After pre-registered exclusions, the final sample comprised $N = 354$ participants (Condition A: $n = 115$; Condition B: $n = 127$; Condition C: $n = 112$). The exclusion funnel proceeded as follows: 471 initial completes → 468 (excluding survey previews by the author during instrument testing) → 451 (excluding pilot responses) → 446 (Prolific platform matching) → 357 (minimum completion time) → 354 (dual attention/comprehension check). Robustness analyses using stricter exclusion criteria (attention-only: $N = 342$; comprehension-only: $N = 289$) produce identical patterns of significance.

3.2.3 Materials

Condition vignettes. Each vignette described WealthPath using matched structural features (0.25% annual fee, $1,000 minimum, SIPC protection, custodial banking) while varying the trust framing. Condition A emphasized fiduciary cues: the platform as a "dedicated digital investment partner" and "steward" with "full discretionary authority" and a commitment to "building a lasting relationship." Condition B emphasized evaluative cues: "systematic strategies," "quantitative analysis," "detailed reports," benchmark comparison, and analytics showing "which decisions are working and why." Condition C provided a bare product description with no emphasis in either direction.

Critically, all three vignettes described the same technological agent — no human advisors were introduced in any condition. This addresses a confound in earlier algorithm trust research where fiduciary framing was bundled with human agency.

Adverse event. All participants received an identical notification: their WealthPath portfolio declined 8.7% during a quarter in which the S&P 500 declined 8.0%, producing a loss of $870 on a $10,000 investment. The 70-basis-point underperformance was designed to be attributionally ambiguous — enough to register as underperformance for attentive evaluators, but close enough to the benchmark to be dismissible by delegative trustors.

3.2.4 Measures

All items used a 7-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree) unless otherwise noted.

Evaluative trust (2 items, $\alpha = .429$). Items assessed performance-monitoring orientation: "I would want to regularly check WealthPath's performance against market benchmarks" and "I would trust WealthPath primarily because I can verify how it's performing." The low reliability of this composite is a recognized limitation; the scale captures benchmark-monitoring orientation but fails to cohere as a unitary construct. We report results using the composite but note the attenuation of evaluative trust effects throughout.

Delegative trust (3 items, $\alpha = .807$). Items assessed willingness to grant discretionary authority: "I would be comfortable giving WealthPath full discretion to adjust my portfolio as it sees fit"; "I would be willing to let WealthPath make investment decisions without needing to understand each one"; and "Even if I could review every decision WealthPath makes, I probably wouldn't bother."

Betrayal (1 item). "After receiving WealthPath's notification about the decline, I feel a sense of betrayal."

Disappointment (1 item). "After receiving WealthPath's notification about the decline, I feel disappointed by the results."

Betrayal premium. Computed as the difference between betrayal and disappointment scores (Item 6a minus Item 6b), operationalizing Holton's (1994) reactive attitude distinction.

Exit intent (1 item). "My first reaction is to research alternative investment platforms rather than wait for WealthPath's strategy to recover."

Algorithm aversion (1 item). "After this experience, I would prefer to manage my own investments or work with a human financial advisor rather than continue using an algorithm." This item operationalizes Dietvorst et al.'s (2015) specific definition of algorithm aversion as the shift from algorithmic to human judgment, distinct from general exit-seeking behavior.

Comprehension check (1 item, administered immediately after the adverse event). Participants were asked how WealthPath's performance compared to the S&P 500, with four response options. The correct answer (WealthPath performed worse) was selected by 81.6% of the final sample. Comprehension pass/fail was coded as a covariate in all analyses. Following pre-registered protocol, participants who failed the comprehension check were retained in the primary analyses, with robustness checks excluding them.

Financial literacy (3 items). The Lusardi and Mitchell (2014) "Big Three" financial literacy battery: compound interest, inflation, and diversification items. Responses were coded as correct/incorrect and summed to create a 0–3 financial literacy composite.

Attention check (1 item). A directed-choice item administered in the demographic battery prior to condition assignment, instructing participants to select a specific response option. Participants who failed were excluded per the pre-registered protocol.

Demographics. Age, income, investment experience, education, and prior robo-advisor experience, coded as a binary variable derived from a multi-select investment experience item that also captured human advisor use, self-directed brokerage investing, and employer retirement plan participation.

3.2.5 Analytic Strategy

The pre-registered analysis plan specified five hypothesis tests:

H1 (Trust Mode Separation) was tested via one-way ANOVA comparing evaluative and delegative trust composites across conditions, with planned contrasts between Conditions A and B.

H2 (Betrayal Premium) was tested via one-way ANOVA comparing the betrayal premium (6a minus 6b) across conditions, with the prediction that Condition A would produce the largest premium.

H3 (Algorithm Aversion Mechanism) was tested via hierarchical multiple regression predicting algorithm aversion (Item 7b) from evaluative trust, delegative trust, and condition dummies, with the pre-registered prediction that evaluative trust would be the significant predictor.

H4 (Delegative Resilience) was tested via hierarchical multiple regression predicting exit intent (Item 7a) from evaluative trust, delegative trust, and condition dummies.

H5 (Financial Literacy Interaction) was tested via moderated regression including a financial literacy × condition interaction term predicting delegative trust.

All analyses were conducted using the pre-registered exclusion criteria. Robustness analyses using stricter attention-only ($N = 342$) and comprehension-only ($N = 289$) exclusion thresholds are reported alongside primary results.

4 CFPB Complaint Analysis: Results

The CFPB analysis provides naturalistic evidence for dimensional separability. Where the Prolific experiment tests whether trust dimensions predict distinct post-shock responses in a controlled setting, the CFPB analysis asks whether evaluative and delegative trust are distinguishable in the language of real consumer complaints — filed by people with actual financial grievances, not experimental participants responding to vignettes. The two analyses are methodologically independent: different populations, different product domains, different measurement approaches, and different analytic strategies. Neither depends on the other. This is the triangulation.

4.1 Sample and Classification Procedure

We sampled 9,000 consumer complaint narratives from the CFPB database, equally distributed across three product categories: mortgage ($n = 3{,}000$), checking and savings ($n = 3{,}000$), and credit card ($n = 3{,}000$). These categories were selected to vary in the degree to which the consumer–institution relationship involves delegation of discretionary authority. Mortgage relationships are long-duration, high-stakes, and involve ongoing servicing decisions that consumers cannot easily monitor — conditions that our theoretical framework identifies as conducive to delegative trust. Credit card relationships, by contrast, are largely transactional: the consumer evaluates terms, monitors statements, and assesses performance on observable metrics. Checking and savings accounts fall between these poles.

Each narrative was independently classified by two large language models — Claude 3.5 Sonnet (Anthropic) and GPT-4o (OpenAI) — using identical prompts that defined the evaluative and delegative trust constructs and asked for (a) a categorical classification (evaluative, delegative, or unclassifiable), (b) continuous scores on each dimension (0–1), and (c) a confidence rating (0–1). Of the 9,000 narratives, Sonnet produced valid classifications for 8,991 (99.9%) and GPT-4o for 8,943 (99.4%). Dual-model analyses use the $N = 8{,}943$ narratives classified by both models.

4.2 Cross-Product Delegative Gradient

The central prediction is that delegative trust indicators should vary across product categories in a theoretically predictable way: highest for mortgages (long-term, opaque servicing), intermediate for checking and savings (ongoing relationship, moderate complexity), and lowest for credit cards (transactional, transparent terms). Evaluative trust indicators, by contrast, should be relatively uniform, since all three product types involve assessable performance dimensions.

The predicted monotonic ordering — mortgage > checking/savings > credit card — holds for delegative indicators across all three analysis variants: Sonnet-only, GPT-4o-only, and concordant cases (Table 1). The gradient is robust. For mortgage versus credit card, the delegative score difference is statistically signif-

Table 1: Cross-product delegative gradient: classification proportions and continuous scores by product category

	Mortgage ($n = 3{,}000$)	Checking/Savings ($n = 3{,}000$)	Credit Card ($n = 3{,}000$)
Panel A: Delegative classification proportion			
Sonnet	21.1%	16.4%	12.3%
GPT-4o	19.8%	13.2%	8.8%
Concordant only	16.7%	11.9%	7.1%
Panel B: Mean delegative score (SD)			
Sonnet	0.335 (0.249)	0.300 (0.228)	0.263 (0.210)
GPT-4o	0.411 (0.219)	0.371 (0.194)	0.329 (0.169)
Concordant avg.	0.347 (0.220)	0.310 (0.196)	0.269 (0.163)
Panel C: Mean evaluative score (SD)			
Sonnet	0.718 (0.210)	0.708 (0.220)	0.711 (0.233)
GPT-4o	0.724 (0.195)	0.705 (0.207)	0.731 (0.212)
Concordant avg.	0.749 (0.188)	0.736 (0.204)	0.749 (0.212)

Note. Each product category comprises $n = 3{,}000$ sampled narratives. Sonnet = Claude 3.5 Sonnet (Anthropic); GPT-4o = GPT-4o (OpenAI). Concordant cases are narratives where both models assign the same categorical classification ($N = 7{,}369$). Delegative and evaluative scores range from 0 to 1. Concordant avg. reports the mean of Sonnet and GPT-4o continuous scores for concordant cases.

icant and of moderate size: Sonnet $d = 0.313$ ($U = 5{,}413{,}026$, $p < .001$); GPT-4o $d = 0.419$ ($U = 5{,}545{,}242$, $p < .001$); concordant averaged scores $d = 0.399$ ($U = 3{,}882{,}984$, $p < .001$).

Evaluative scores, by contrast, are essentially flat across products. The mortgage–credit card difference in evaluative scores is negligible: Sonnet $d = 0.030$; GPT-4o $d = -0.031$; concordant $d = 0.001$ (n.s., $p = .145$). This asymmetry is precisely what dimensional separability predicts. If the two trust modes were a single underlying construct — "complaint intensity" or "relationship quality" — we would expect both dimensions to co-vary across product types. Instead, only delegative trust tracks the theoretically relevant product dimension (relationship opacity and duration), while evaluative trust remains stable.

4.3 Dual-Model Concordance

A classification scheme is only useful if it can be applied reliably.

Table 2: Confidence-stratified inter-model concordance ($N = 8{,}943$)

Confidence tercile	Range	N	Agreement	Cohen's κ
Low	0.51–0.78	3,759	68.8%	0.356
Medium	0.78–0.83	3,124	87.7%	0.652
High	0.85–0.97	2,060	99.2%	0.970
Overall	—	8,943	82.4%	0.556

Note. Confidence is the mean of Sonnet and GPT-4o confidence ratings for each narrative. Tercile boundaries are based on the distribution of mean confidence scores. Agreement = percentage of narratives receiving the same categorical classification (evaluative, delegative, or unclassifiable) from both models. Cohen's κ corrects for chance agreement.

Overall categorical agreement is 82.4% ($\kappa = 0.556$, moderate; Table 2). This headline figure, however, masks a sharp confidence–concordance gradient. When both models assign high confidence, agreement

reaches 99.2% ($\kappa = 0.970$, almost perfect). When confidence is low, agreement drops to 68.8% ($\kappa = 0.356$). This pattern is theoretically informative: the narratives that are difficult for classifiers are the ambiguous cases where trust mode is genuinely unclear, not cases where the classification scheme is arbitrary.

Continuous scores show strong cross-model convergence. Sonnet and GPT-4o delegative scores correlate at $r = .703$ (Spearman $\rho = .738$, $p < .001$); evaluative scores at $r = .680$ ($\rho = .713$, $p < .001$). Both models independently recover the same latent structure in the complaint narratives.

4.4 Moral Foundations Validation

As an independent check on construct validity, we scored all narratives using the extended Moral Foundations Dictionary (eMFD) and compared moral language across trust classifications. The theoretical prediction is specific: delegative trust violations should activate relational moral foundations — particularly loyalty (the foundation most closely associated with trust betrayal) — more than evaluative trust violations.

Narratives classified as delegative scored significantly higher than evaluative narratives on all five moral foundations, with the largest effects for loyalty ($d = 0.267$, $p < .001$) and sanctity ($d = 0.263$, $p < .001$), followed by care ($d = 0.235$, $p < .001$). The loyalty result is particularly important: in the Moral Foundations framework, the loyalty foundation tracks in-group betrayal and allegiance — precisely the relational vocabulary our theory associates with delegative trust violation.

Table 3: eMFD correlations with trust scores: dual-model comparison (Spearman ρ)

	Delegative score		Evaluative score	
Foundation	Sonnet	GPT-4o	Sonnet	GPT-4o
Care	.099***	.129***	−.112***	−.120***
Fairness	.069***	.106***	−.086***	−.103***
Loyalty	.118***	.158***	−.108***	−.128***
Authority	.035**	.074***	−.062***	−.078***
Sanctity	.108***	.155***	−.119***	−.139***

Note. $N = 8{,}943$. Moral foundation scores derived from the extended Moral Foundations Dictionary (eMFD). Delegative and evaluative scores are continuous (0–1) as assigned by each model.
* $p < .05$, ** $p < .01$, *** $p < .001$.

Correlation patterns reinforce this interpretation (Table 3). Delegative scores correlate positively with all foundations, most strongly with loyalty ($\rho = .118$, $p < .001$). Evaluative scores correlate *negatively* with all foundations, most strongly with sanctity ($\rho = -.119$, $p < .001$) and care ($\rho = -.112$, $p < .001$). These opposing sign patterns hold for both Sonnet and GPT-4o classifications, providing further evidence that the two models are recovering the same construct distinction.

4.5 Human Validation

To assess whether the AI classifications capture distinctions recognizable to a human coder, I independently classified a confidence-stratified subsample of 100 narratives. Stratification ensured representation across confidence levels: 30 low-confidence, 40 medium, and 30 high-confidence narratives (based on Sonnet confidence bins).

Categorical agreement was fair by conventional benchmarks: $\kappa = 0.264$ for human–Sonnet and $\kappa = 0.361$ for human–GPT-4o. The two AI models agreed more with each other ($\kappa = 0.590$) than either did with

the human coder. However, the confidence-stratified pattern is dramatic and informative. At high confidence, human–Sonnet agreement reaches 93.3% and human–GPT-4o reaches 90.0%. At low confidence, agreement drops to 30.0% and 50.0%, respectively.

The primary source of disagreement is systematic and interpretable: the human coder classified 56% of narratives as unclassifiable, compared to 21% (Sonnet) and 33% (GPT-4o). The most common disagreement pattern — 23 of 51 disagreement cases — was both AI models classifying a narrative as evaluative where the human coded it as unclassifiable. This reflects a higher threshold for inferring trust language in the human coding: factual complaints that describe service failures without explicit evaluative language (e.g., "they put a hold on my account and I can't access it") were coded unclassifiable by the human but evaluative by the models, which inferred frustration with performance from the complaint structure.

Continuous score correlations tell a more encouraging story. Human and model delegative scores correlate moderately (human–Sonnet $r = .488$, $p < .001$; human–GPT-4o $r = .511$, $p < .001$), and both are substantially lower than the inter-model correlation ($r = .776$). Evaluative score correlations follow the same hierarchy (human–Sonnet $r = .308$; human–GPT-4o $r = .402$; inter-model $r = .779$). The models agree with each other more than with the human, but the human–model agreement is significant and in the expected direction, particularly at high confidence.

We interpret this pattern as follows. The dual-model approach functions as an automated coding system with known biases (over-attribution of evaluative intent in factual narratives). The high-confidence subset, where all three raters converge, represents the clearest cases of each trust mode. The analyses reported above are robust to restricting to concordant cases (see Table 1, concordant rows), which partially addresses this concern.

4.6 Dimensional Relationship

Our theoretical framework characterizes evaluative and delegative trust as independent dimensions. In the Prolific experiment, the two composites correlate at $r = .246$ (6% shared variance; Table 6), consistent with dimensional separability. The CFPB data present a more complex picture.

Across the full sample, delegative and evaluative continuous scores are moderately negatively correlated (Sonnet $r = -.592$; GPT-4o $r = -.574$; concordant averaged $r = -.561$). This is substantially stronger than the Prolific correlation and in the opposite direction. Three features of the data suggest this reflects properties of the CFPB sample rather than the constructs themselves.

First, the correlation varies substantially across product types: mortgage $r = -.807$, checking/savings $r = -.589$, credit card $r = -.400$ (Sonnet). If the negative correlation reflected a true inverse relationship between the constructs, we would expect it to be stable across contexts. Instead, it tracks product complexity — strongest where the complaint space is most constrained (mortgages, where nearly every complaint involves either servicing failures or relationship violations) and weakest where complaints are more heterogeneous (credit cards).

Second, the scatter plot (Figure 1) reveals that the correlation is partly driven by clustering at classification-typical regions rather than a smooth negative slope. Narratives cluster in a high-evaluative/low-delegative region and a high-delegative/moderate-evaluative region, with relatively sparse representation in the joint-high quadrant. This is consistent with the classification mechanics: when a narrative strongly expresses one trust mode, the classifier assigns lower scores to the other mode, producing a mechanical negative correlation within the classification system.

Third, and most importantly, the CFPB sample is censored: it consists entirely of people who have filed

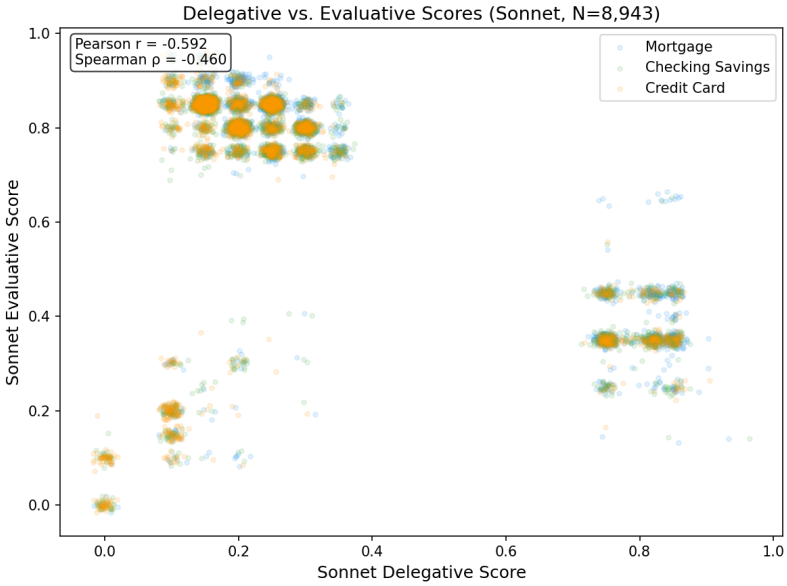

Figure 1: Delegative versus evaluative trust scores (Sonnet classifications, $N = 8{,}943$). Points are colored by product type. The moderate negative correlation ($r = -.592$) is partly driven by clustering at classification-typical regions rather than a smooth inverse relationship.

formal complaints. These consumers have already experienced a breakdown in their relationship with the institution. In the language of our framework, the sample is heavily weighted toward cases where at least one trust mode has been violated. In a general population, individuals may hold high levels of both evaluative and delegative trust simultaneously (as the Prolific data suggest). In the CFPB sample, the complaint context compresses the joint distribution toward the violated region of the trust space.

We therefore treat the negative correlation as an informative feature of the complaint context — consistent with the theory's prediction that trust violations are mode-specific — rather than evidence against dimensional independence. The stronger test of independence comes from the Prolific general-population sample, where the weak positive correlation ($r = .246$, 6% shared variance) supports separability.

4.7 Robustness: Narrative Length

Delegative narratives are longer than evaluative narratives across all product types (e.g., mortgage: delegative $M = 2{,}144$ characters vs. evaluative $M = 1{,}590$). This raises a potential confound: perhaps longer narratives simply provide more text for classifiers to identify relational cues, and the delegative classification is partly an artifact of narrative length.

To address this, we regressed Sonnet delegative scores on log-transformed character count and product category dummies. Length is a significant predictor ($b = 0.079$, $p < .001$). However, the cross-product delegative gradient retains 74.3% of its magnitude after controlling for length (raw gradient: 0.071; residualized gradient: 0.053).[2] The theoretically meaningful between-product variation in delegative trust cannot be explained by narrative length alone.

[2]Gradient values computed from unrounded mean delegative scores.

4.8 Exploratory: Company-Level Variation

As an exploratory analysis, we examined whether specific mortgage companies generate disproportionately more delegative complaints. Among the 15 companies with 50 or more mortgage narratives, delegative classification proportions ranged from 3.6% to 37.8% — a spread of 34.2 percentage points. This variation suggests that institutional behavior may drive the trust mode activated in consumer complaints, a hypothesis consistent with our framework's claim that delegative trust responds to relational rather than performance signals. We return to this finding in Section 6.

4.9 Summary

The CFPB analysis provides naturalistic evidence for three aspects of our theoretical framework. First, the cross-product delegative gradient — consistent across two independent AI classifiers and robust to narrative length controls — demonstrates that trust modes are separable in real consumer behavior and track theoretically predicted product characteristics. Second, the eMFD analysis shows that delegative complaints activate relational moral vocabulary, particularly loyalty, at higher rates than evaluative complaints. Third, the dual-model approach, validated against human coding at high confidence levels, provides a reliable method for identifying trust modes in unstructured text at scale.

The CFPB database does not cover investment advisory services, which fall under SEC and FINRA jurisdiction rather than CFPB oversight. This means we cannot directly study robo-advisor complaints. We treat this jurisdictional boundary as a triangulation feature rather than a limitation: the CFPB analysis covers banking products using naturalistic complaint data, while the Prolific experiment covers algorithmic investment advice using controlled vignettes. The theoretical framework's predictions hold across both domains, different populations, and different measurement approaches.

5 Prolific Experiment: Results

5.1 Sample and Exclusions

We recruited 471 responses through Prolific. Table 4 reports the pre-registered exclusion pipeline. After removing survey previews (3), pre-registered pilot responses (17), and matching to Prolific's approved submissions, 446 responses remained. We then applied two pre-registered exclusion criteria: completion time under 120 seconds removed 89 responses (19.9%), and the dual failure criterion — failing both the instructed attention check and the comprehension check — removed an additional 3 responses. The final analytical sample comprised 354 participants (Condition A: $n = 115$; Condition B: $n = 127$; Condition C: $n = 112$).

Table 4: Sample exclusion pipeline

Stage	*N* Remaining	*N* Excluded	Reason
Qualtrics total	471	—	Raw export
Remove previews	468	3	Survey preview responses
Remove pilots	451	17	Pre-registered pilot exclusion
Prolific matching	446	5	Unmatched to Prolific approved submissions
Speed exclusion	357	89	Completion time $<$ 120 seconds
Dual-check exclusion	354	3	Failed both attention and comprehension checks

The speed exclusion rate warrants comment. Approximately one in five participants completed the survey in under two minutes, a duration insufficient to read the vignette, process the performance notification, and respond thoughtfully to the dependent measures. Among the 357 participants who met the speed threshold, only 3 failed the dual attention-comprehension criterion, suggesting that the speed filter effectively separated engaged from disengaged respondents.

A condition balance check revealed a significant age imbalance: participants in Condition C were older ($M = 41.2$) than those in Condition A ($M = 37.0$) and Condition B ($M = 37.5$), $F(2, 351) = 4.15$, $p = .017$. All other covariates — financial literacy, gender, education, income, and robo-advisor experience — were balanced across conditions. Age is included as a covariate in all regression models.

5.2 Descriptive Statistics

Table 5: Means and standard deviations of dependent variables by experimental condition

	Condition A ($n = 115$)	Condition B ($n = 127$)	Condition C ($n = 112$)	Overall ($N = 354$)
Evaluative trust ($\alpha = .43$)	5.43 (1.01)	5.28 (1.12)	5.19 (1.01)	5.30 (1.05)
Delegative trust ($\alpha = .81$)	3.84 (1.39)	4.04 (1.48)	3.86 (1.39)	3.92 (1.42)
Betrayal	4.23 (1.61)	4.02 (1.71)	4.24 (1.61)	4.16 (1.64)
Disappointment	5.84 (1.16)	5.71 (1.36)	5.71 (1.18)	5.75 (1.24)
Betrayal premium	−1.62 (1.57)	−1.69 (1.65)	−1.46 (1.63)	−1.59 (1.62)
General exit	4.68 (1.60)	4.57 (1.72)	4.76 (1.58)	4.67 (1.64)
Algorithm aversion	5.03 (1.53)	4.65 (1.58)	4.95 (1.43)	4.86 (1.52)

Note. All items measured on 7-point Likert scales (1 = Strongly Disagree, 7 = Strongly Agree). Betrayal premium = Betrayal minus Disappointment. Condition A = Fiduciary/Relationship framing; Condition B = Performance/Analytics framing; Condition C = Neutral control. Cronbach's α reported for composite measures.

Table 5 presents means and standard deviations for all dependent variables by condition. Two patterns merit attention. First, evaluative trust ($M = 5.30$, $SD = 1.05$) was substantially higher than delegative trust ($M = 3.92$, $SD = 1.42$) across all conditions. Participants were generally willing to monitor and evaluate the algorithm's performance but ambivalent about granting it discretionary authority — a pattern consistent with the framework's prediction that the default psychological posture toward algorithms is evaluative. Second, disappointment ($M = 5.75$) exceeded betrayal ($M = 4.16$) in every condition, producing a consistently negative betrayal premium ($M = -1.59$). After an adverse event, participants' dominant reactive attitude was disappointment at the outcome rather than betrayal by the agent, again consistent with evaluative rather than delegative trust governing the human–algorithm relationship.

The delegative trust composite showed good internal consistency (Cronbach's $\alpha = .807$). The evaluative trust composite, comprising only two items, showed poor internal consistency ($\alpha = .429$). We discuss implications of this measurement limitation below and in Section 6.

Table 6 presents bivariate correlations among the trust composites and dependent variables. The evaluative and delegative trust composites are positively correlated ($r = .246$, $p < .001$), confirming that they are related constructs — as any two trust measures should be — while sharing only approximately 6% of variance. The two composites thus capture substantially distinct dimensions of the participant's orientation toward the algorithm. Their downstream profiles reinforce this conclusion. Delegative trust is negatively correlated with algorithm aversion ($r = -.313$, $p < .001$) and general exit ($r = -.155$, $p < .01$), but uncorrelated with betrayal ($r = .061$, n.s.). Evaluative trust shows a largely flat profile across dependent

variables, consistent with both its lower reliability and the descriptive observation that evaluative engagement is near-uniformly high across the sample. Among the post-shock dependent variables, general exit and algorithm aversion are strongly correlated ($r = .690$, $p < .001$), reflecting their shared orientation toward abandoning the algorithm, though they differ in specificity: general exit captures platform-switching intent while algorithm aversion captures the preference for human advice specifically.

Table 6: Bivariate correlations among trust composites and dependent variables

	1	2	3	4	5	6	7
1. Evaluative trust	—						
2. Delegative trust	.246***	—					
3. Betrayal	.054	.061	—				
4. Disappointment	.113*	−.102	.399***	—			
5. Betrayal premium	−.032	.140**	.712***	−.359***	—		
6. General exit	−.015	−.155**	.578***	.406***	.278***	—	
7. Algorithm aversion	−.109*	−.313***	.515***	.420***	.202***	.690***	—

Note. $N = 354$. Evaluative trust = mean of Items 1–2; Delegative trust = mean of Items 3–5; Betrayal premium = Betrayal minus Disappointment.
* $p < .05$, ** $p < .01$, *** $p < .001$.

5.3 Experimental Manipulation

We predicted that framing the robo-advisor in fiduciary terms (Condition A) would produce higher delegative trust than framing it in performance terms (Condition B), and that the performance framing would produce higher evaluative trust (H1). Neither prediction was supported. One-way ANOVA revealed no significant differences across conditions in delegative trust, $F(2, 351) = 0.75$, $p = .473$, $\eta^2 = .004$, or evaluative trust, $F(2, 351) = 1.50$, $p = .225$, $\eta^2 = .009$ (Table 7). The observed means were directionally opposite to predictions on both measures, though far from significance.

Table 7: One-way ANOVA results for experimental manipulation effects (H1, H2)

Hypothesis	DV	$F(2, 351)$	p	η^2	Predicted	Observed
H1a	Delegative trust	0.75	.473	.004	A > B	B > A (n.s.)
H1b	Evaluative trust	1.50	.225	.009	B > A	A > B (n.s.)
H2	Betrayal premium	0.57	.564	.003	A > B	A > B (n.s.)

Note. Condition A = Fiduciary/Relationship framing; Condition B = Performance/Analytics framing; Condition C = Neutral control. n.s. = not significant at $\alpha = .05$. $N = 354$.

The betrayal premium (H2) was likewise undifferentiated across conditions, $F(2, 351) = 0.57$, $p = .564$, $\eta^2 = .003$. All three conditions produced negative betrayal premiums, indicating that disappointment dominated betrayal regardless of framing.

We interpret the manipulation failure in Section 6. Crucially, the null effect of condition assignment does not bear on the validity of the trust constructs as individual-difference measures. The primary predictions of the evaluative–delegative framework concern the downstream consequences of trust modes, not their experimental inducibility, and we test these predictions in the analyses that follow.

5.4 Delegative Trust and Post-Shock Resilience

Our framework predicts that delegative trust — granting the advisor discretionary authority with acceptance of opacity — should buffer against exit-seeking behavior after an adverse event, because the delegative trustor's relationship with the advisor is not contingent on any particular performance outcome. H4 tests this prediction directly.

Table 8 reports the full regression model predicting general exit intent. The overall model is significant, $F(9, 344) = 2.18$, $p = .023$, $R^2 = .054$. As predicted, delegative trust negatively predicts exit intent ($B = -0.206$, $SE = 0.064$, $\beta = -0.179$, $p = .001$, 95% CI $[-0.33, -0.08]$). Each unit increase in delegative trust on the seven-point scale is associated with a 0.21-point decrease in the intention to research alternative platforms. Evaluative trust, by contrast, shows no predictive relationship with exit intent ($B = 0.037$, $p = .667$). The two trust dimensions thus make divergent predictions about post-shock behavior: one predicts resilience, the other is inert.

Table 8: OLS regression predicting general exit intent (H4)

Predictor	B	SE	β	t	p	95% CI
Delegative trust	−0.206**	**0.064**	−0.179	−3.23	**.001**	**[−0.33, −0.08]**
Evaluative trust	0.037	0.087	0.024	0.43	.667	[−0.13, 0.21]
Condition A (Fiduciary)	−0.056	0.218	−0.016	−0.26	.797	[−0.48, 0.37]
Condition B (Performance)	−0.132	0.211	−0.039	−0.62	.532	[−0.55, 0.28]
Age	0.005	0.007	0.040	0.74	.460	[−0.01, 0.02]
Financial literacy	−0.251	0.134	−0.106	−1.87	.062	[−0.51, 0.01]
Robo-advisor experience	0.242	0.266	0.049	0.91	.365	[−0.28, 0.76]
Comprehension pass	0.616**	0.232	0.146	2.65	.008	[0.16, 1.07]
Attention pass	0.577	0.478	0.064	1.21	.228	[−0.36, 1.52]
$R^2 = .054$, Adjusted $R^2 = .029$, $F(9, 344) = 2.18$, $p = .023$						

Note. Reference category for condition is C (Neutral control). Robo-advisor experience coded as 1 = has used a robo-advisor. Comprehension pass = correctly identified WealthPath underperformance. Attention pass = selected instructed response. $N = 354$. * $p < .05$, ** $p < .01$, *** $p < .001$.

Two additional findings are worth noting. Comprehension check pass positively predicts exit intent ($B = 0.616$, $p = .008$): participants who correctly identified the performance shortfall were more likely to seek alternatives. This pattern is consistent with evaluative processing of the adverse event — the negative signal was encoded and acted upon — though the effect is captured by the check variable rather than the evaluative trust composite, a point to which we return below. Financial literacy approaches significance as a negative predictor ($B = -0.251$, $p = .062$), suggesting that higher-literacy participants may be more tolerant of short-term underperformance.

The result is robust to alternative exclusion criteria. Under strict attention exclusion ($N = 342$), the delegative trust coefficient is $B = -0.207$, $p = .001$. Under strict comprehension exclusion ($N = 289$), it is $B = -0.161$, $p = .021$. Direction and significance are unchanged.

5.5 Delegative Trust and Algorithm Aversion

H3 tested whether the trust dimensions predict algorithm aversion — the preference for a human advisor or self-management over the algorithm after the adverse event. We predicted that evaluative trust would drive algorithm aversion and that delegative trust would not. The results reversed this prediction in a

theoretically informative way.

Table 9 reports the full model, which is highly significant, $F(9, 344) = 6.82$, $p < .001$, $R^2 = .151$, and represents the strongest overall fit of any regression in the experimental analyses. Delegative trust is the dominant predictor of algorithm aversion ($B = -0.326$, $SE = 0.056$, $\beta = -0.304$, $p < .001$, 95% CI $[-0.44, -0.22]$). Participants who scored higher on willingness to grant the algorithm discretionary authority were substantially less likely to prefer a human advisor after the adverse event. Evaluative trust, by contrast, was not significant ($B = -0.039$, $p = .614$).

Table 9: OLS regression predicting algorithm aversion (H3)

Predictor	B	SE	β	t	p	95% CI
Delegative trust	−0.326***	**0.056**	−0.304	−5.79	< .001	**[−0.44, −0.22]**
Evaluative trust	−0.039	0.077	−0.027	−0.50	.614	[−0.19, 0.11]
Condition A (Fiduciary)	0.073	0.192	0.023	0.38	.703	[−0.30, 0.45]
Condition B (Performance)	−0.250	0.186	−0.079	−1.35	.179	[−0.62, 0.12]
Age	−0.006	0.006	−0.051	−1.01	.315	[−0.02, 0.01]
Financial literacy	−0.191	0.118	−0.086	−1.62	.107	[−0.42, 0.04]
Robo-advisor experience	−0.215	0.234	−0.047	−0.92	.358	[−0.68, 0.25]
Comprehension pass	0.584**	0.204	0.149	2.86	.004	[0.18, 0.99]
Attention pass	1.179**	0.421	0.141	2.80	.005	[0.35, 2.01]
$R^2 = .151$, Adjusted $R^2 = .129$, $F(9, 344) = 6.82$, $p < .001$						

Note. Reference category for condition is C (Neutral control). Robo-advisor experience coded as 1 = has used a robo-advisor. Comprehension pass = correctly identified WealthPath underperformance. Attention pass = selected instructed response. $N = 354$.
* $p < .05$, ** $p < .01$, *** $p < .001$.

This finding has a straightforward interpretation within the evaluative–delegative framework. The factor that protects against algorithm aversion is not a favorable performance assessment — it is the willingness to cede decisional authority. Participants who had granted the algorithm discretion, who had accepted opacity and declined to second-guess every decision, maintained that posture through an adverse event. Those who had not granted such discretion showed no corresponding resilience. This is dimensional separability operating in the predicted direction: the two trust modes engage different psychological mechanisms, and only the delegative mechanism predicts who stays with the algorithm after failure.

As in H4, the check variables are informative. Both comprehension pass ($B = 0.584$, $p = .004$) and attention pass ($B = 1.179$, $p = .005$) positively predict algorithm aversion. Participants who were paying attention and who correctly encoded the performance shortfall were more, not less, likely to want a human advisor. This pattern is consistent with evaluative processing — the participants who received and encoded the negative signal were the ones who responded by seeking alternatives — but the evaluative trust composite does not capture this effect. The poor internal consistency of the evaluative trust composite ($\alpha = .429$) likely explains the discrepancy: the comprehension and attention check variables are acting as proxies for evaluative engagement that the two-item scale measures inadequately. This is a limitation of the present study; we discuss it further in Section 6, but note here that the limitation biases against finding evaluative trust effects and does not threaten the delegative trust findings.

This result also held under both robustness checks. Strict attention exclusion ($N = 342$): delegative trust $B = -0.322$, evaluative trust $B = -0.058$ (n.s.). Strict comprehension exclusion ($N = 289$): delegative trust $B = -0.316$, evaluative trust $B = -0.047$ (n.s.).

5.6 Financial Literacy Interaction

H5 predicted that financial literacy would moderate the effect of condition on delegative trust, with high-literacy participants showing a larger difference between Conditions A and B. The interaction was not significant (Condition A $\times$ high literacy: $B = 0.072$, $p = .869$; Condition B $\times$ high literacy: $B = -0.109$, $p = .797$). This null follows from the manipulation failure: where condition assignment does not produce main effects on delegative trust, an interaction with literacy is unlikely to emerge.

One finding from the H5 model merits attention independent of the interaction. Prior robo-advisor experience was the strongest predictor of delegative trust in the model ($B = 0.729$, $p = .001$). Participants who had previously used a robo-advisor scored nearly three-quarters of a point higher on the delegative trust composite, regardless of experimental condition or financial literacy. This is a substantial effect that dwarfs any condition-level variation and is consistent with the philosophical account of trust developed in Section 2: trust postures are learned through repeated encounters rather than installed by informational framing. We return to this finding in Section 6.

5.7 Exploratory Analyses

Three pre-registered exploratory analyses are reported here. These were designated as exploratory in the pre-registration and should not be treated as confirmatory.

We predicted that Condition A would produce higher variance in delegative trust than the other conditions (E1), reflecting polarization between participants who can and cannot extend delegative trust to algorithms. This was not supported: Levene's test was not significant, $F(2, 351) = 0.32$, $p = .728$, and variances were similar across conditions (A: 1.94; B: 2.18; C: 1.92).

We predicted that comprehension check failures would show higher delegative trust and lower evaluative trust than passes (E2), on the theory that delegative trustors encode performance signals less carefully. The predicted directions emerged on both measures — failures showed slightly higher delegative trust ($M = 3.97$ vs. 3.90) and lower evaluative trust ($M = 5.11$ vs. 5.34) — but neither difference reached significance (delegative: $t = -0.33$, $p = .739$; evaluative: $t = 1.60$, $p = .110$). Comprehension failure rates did not differ meaningfully by condition (A: 19.1%; B: 18.9%; C: 17.0%; Table 10).

Table 10: Comprehension check performance by experimental condition

Condition	Pass *N* (%)	Fail *N* (%)	Total
A (Fiduciary)	93 (80.9%)	22 (19.1%)	115
B (Performance)	103 (81.1%)	24 (18.9%)	127
C (Neutral)	93 (83.0%)	19 (17.0%)	112
Total	289 (81.6%)	65 (18.4%)	354

Note. Comprehension check asked participants to identify WealthPath's performance relative to the S&P 500. Correct answer: WealthPath performed worse than the market.

We tested whether prior robo-advisor experience moderated the relationship between condition and each dependent variable (E3). Two of fourteen interactions reached significance: Condition A $\times$ robo-experience on disappointment ($B = 1.174$, $p = .022$), and Condition B $\times$ robo-experience on the betrayal premium ($B = -1.313$, $p = .035$). Among participants with robo-advisor experience, fiduciary framing produced greater disappointment after the adverse event, while performance framing produced a more

negative betrayal premium — that is, a stronger dominance of disappointment over betrayal. Both patterns are consistent with the framework: experienced users who are told the algorithm is their fiduciary may have activated higher expectations, while experienced users primed with performance framing process the shortfall as a pure evaluative failure. However, with fourteen interaction tests conducted at $\alpha = .05$, approximately 0.7 significant results are expected by chance. These two interactions modestly exceed chance expectation but would not survive correction for multiple comparisons. We treat them as suggestive and hypothesis-generating.

5.8 Summary

The experimental manipulation did not differentiate trust modes across conditions, but the evaluative–delegative distinction proves consequential when measured as a continuous individual difference. The two trust composites share only 6% of variance ($r = .246$) and show divergent downstream profiles (Table 6). Delegative trust is the primary predictor of resilience to both exit-seeking (H4: $\beta = -0.179$, $p = .001$) and algorithm aversion (H3: $\beta = -0.304$, $p < .001$) following an adverse event. Evaluative trust predicts neither outcome, though this null must be interpreted cautiously given the low reliability of the two-item composite. The trust dimensions thus make divergent and specific predictions about post-shock behavior, supporting the core theoretical claim of dimensional separability. These findings complement the CFPB analysis reported in Section 4, which provides naturalistic evidence for the same constructs; we discuss the triangulation in Section 6.

6 Discussion

The results reported above provide convergent evidence from two independent methods that evaluative and delegative trust are separable dimensions with distinct behavioral signatures. They also reveal important boundary conditions on the framework — most notably, the failure to manipulate trust modes through vignette framing — that are themselves theoretically informative. We organize the discussion around four themes: what the manipulation failure teaches, the pattern of dimensional separability across methods, the reconciliation of the algorithm aversion literature, and the practical implications of delegative foreclosure.

6.1 What the Manipulation Failure Teaches

The most straightforward prediction of our framework — that fiduciary framing would elevate delegative trust and performance framing would elevate evaluative trust (H1) — was not supported. All between-condition comparisons were nonsignificant (all $p > .22$, all $\eta^2 < .01$). The manipulation did not shift participants between trust modes.

We do not treat this as evidence against the evaluative/delegative distinction. The distinction is supported by the individual-differences analyses (H3, H4) and by the CFPB naturalistic data, neither of which depends on the manipulation. But the failure requires explanation, and the explanation is substantive rather than merely methodological.

Hardin's (1993) street-level epistemology provides the key. If trust postures are learned Bayesian priors — dispositions the trustor brings to the encounter based on accumulated experience with a category of interaction — then a brief vignette describing a robo-advisor in fiduciary terms cannot install a delegative posture in participants who have not learned, through experience, that algorithmic systems can be trusted

with discretionary authority. The manipulation asks participants to *imagine* delegating judgment; it cannot provide the experiential grounding that delegative trust requires.

The robo-advisor experience predictor supports this interpretation directly. Prior experience with robo-advisory platforms was the strongest predictor of delegative trust in the sample ($B = 0.729$, $p = .001$), substantially stronger than any experimental manipulation. Participants who had actually used algorithmic advisors — who had the experiential encounters Hardin's framework identifies as necessary for trust formation — exhibited higher delegative trust regardless of condition assignment. This is not a confound; it is the mechanism. Delegative trust toward algorithms forms through repeated encounters in which vulnerability is experienced as safe, not through institutional framing alone.

This finding has implications for both research and practice. Experimentally, it suggests that single-session vignette designs — the overwhelming majority of algorithm aversion experiments use such designs (Chacon et al., 2022), with rare longitudinal exceptions (e.g., Lee et al., 2026) — may be structurally unable to study delegative trust formation, because they cannot provide the experiential history that delegative trust requires. Longitudinal designs, in which participants interact with algorithmic advisors over multiple sessions and encounter both favorable and adverse outcomes, may be necessary. For practitioners, the implication is that onboarding framing is insufficient: fiduciary language in terms of service or welcome screens is unlikely to build delegative trust without a sustained relationship that gives the framing experiential content.

6.2 Dimensional Separability Across Methods

The core theoretical claim — that evaluative and delegative trust are separable dimensions responding to different signal classes (Claim 1) — is supported by two independent lines of evidence.

In the CFPB analysis, the cross-product delegative gradient provides the cleanest test. Delegative trust indicators decline monotonically from mortgage (21.1%) to checking/savings (16.4%) to credit card (12.3%), tracking the theoretically predicted dimension of relationship opacity and duration. Evaluative trust indicators are flat across the same products (Sonnet $d = 0.030$ for mortgage vs. credit card). This asymmetric pattern — one dimension varying while the other holds constant — is the signature of dimensional separability. It is robust across two independent AI classifiers and survives control for narrative length (74.3% of the gradient retained).

In the Prolific experiment, separability appears in the predictive structure. Delegative trust predicts both exit intent (H4: $B = -0.206$, $p = .001$) and algorithm appreciation — that is, continued preference for algorithmic advice despite an adverse event (H3: $B = -0.326$, $p < .001$). Evaluative trust predicts neither (both $p > .6$). The two dimensions are not merely distinguishable in language — they predict different behaviors in a controlled post-shock context. The correlation between the two trust composites ($r = .246$, 6% shared variance) further supports separability: related but far from redundant.

The triangulation is strengthened by the methodological independence of these findings. The CFPB analysis uses naturalistic complaint text, AI-based classification, and a between-product design across banking products. The Prolific experiment uses self-report measures, a between-subjects vignette design, and an algorithmic investment context. The two analyses share no participants, no measurement instruments, no product domain, and no analytic strategy. That both converge on dimensional separability — through different operationalizations, in different populations, across different financial products — constitutes substantially stronger evidence than either alone.

6.3 Reconciling Algorithm Aversion and Appreciation

The evaluative/delegative framework resolves an apparent contradiction in the algorithmic decision-making literature that has persisted for over a decade.

Logg et al.'s (2019) algorithm appreciation and Dietvorst et al.'s (2015) algorithm aversion are not competing findings. Both document evaluative trust operating as our framework predicts: updating monotonically with performance signals. Logg et al.'s participants observed positive signals and increased trust; Dietvorst et al.'s participants observed negative signals and withdrew it. The disagreement in the literature arises from treating these as evidence about fundamentally different human–algorithm relationships, when they are movements along the same evaluative dimension under different evidentiary conditions.

Kormylo et al. (2026) introduce a third finding that completes the picture: betrayal aversion reduces delegation to human experts but not to algorithmic experts. Their explanation — that people do not experience betrayal toward machines — is empirically robust. Our framework provides the structural account of *why* the betrayal response is absent: betrayal is the reactive attitude specific to delegative trust violation, and delegative trust requires the attribution of discretionary authority to the trustee. Under default experimental conditions — including Kormylo et al.'s trust games and delegation tasks — participants approach algorithms as tools to be assessed, not agents to be entrusted with judgment. Delegative trust does not form, so the betrayal response has no ground to stand on.

The three findings thus decompose cleanly within the two-dimensional framework. Logg et al. (2019) document evaluative trust responding to positive signals. Dietvorst et al. (2015) document evaluative trust responding to negative signals. Kormylo et al. (2026) document the absence of delegative trust under default conditions. The literature has been studying variation *within* evaluative trust — appreciating or averting based on observed performance — without recognizing that a second dimension exists.

Our Prolific data add to this picture. The H3 result — that delegative trust, not evaluative trust, is the dominant predictor of sustained algorithm appreciation after an adverse event — demonstrates that the second dimension is not merely a theoretical possibility but an active predictor of behavior. Participants with high delegative trust did not switch to human advice after the performance shock, even though they experienced the same adverse event as participants with low delegative trust. The evaluative dimension, which the existing literature treats as the complete explanation for algorithm aversion, had no predictive power ($p = .614$). This is precisely what the framework predicts: algorithm aversion is an evaluative phenomenon, but sustained appreciation — continued willingness to rely on algorithmic advice — comes from the delegative dimension. Investors who have granted discretionary authority do not revert to performance-based evaluation after a single adverse signal. Convergent evidence comes from Hermann and Alberhasky (2026), who find that affective trust — the dimension closest to what we term delegative — is a substantially stronger mediator than cognitive trust in predicting willingness to follow AI financial advice.

This reinterpretation also generates a novel prediction that none of the three papers can produce independently: if experimental designs create conditions for delegative trust formation toward algorithms — through fiduciary framing combined with sustained interaction — then betrayal-like responses should emerge even toward algorithmic advisors. Our manipulation attempted this and failed, likely because a single vignette cannot provide the experiential grounding required. The prediction remains testable in longitudinal designs.

6.4 Delegative Foreclosure and Design Implications

The concept of delegative foreclosure — introduced in the theoretical framework (Section 2) as the closure of an entire trust mode toward a category of trustee — connects the empirical findings to practical questions about robo-advisor design and financial regulation.

The persistence of algorithm aversion is a well-documented puzzle (see also Belanche et al., 2019; Huang et al., 2026). Dietvorst et al. (2015) find that even showing users evidence of algorithmic superiority does not fully overcome aversion. The standard explanation appeals to evaluative mechanisms: perhaps users weight errors more heavily than successes, or perhaps they hold algorithms to higher standards than human advisors. Our framework offers a structural alternative. For users who have foreclosed delegative trust toward algorithms, the performance evidence is addressing the wrong dimension. Their evaluative channel may be open — they can acknowledge that the algorithm performs well on metrics — but the delegative channel is closed. They cannot bring themselves to grant the algorithm discretionary authority, to accept the vulnerability that delegation requires. No amount of performance evidence is likely to reopen that channel, because the closure is structural, not evidentiary.

This analysis inverts conventional wisdom about robo-advisor design. The industry's dominant strategy — increase transparency, provide performance dashboards, offer explainability tools — is optimized for evaluative trust. It makes the algorithm's decisions legible, verifiable, and assessable against benchmarks. Under our framework, this approach succeeds for users already in evaluative mode (who may appreciate the algorithm per Logg et al., 2019) but fails for users in delegative foreclosure (who need relational repair, not more data) and actively undermines nascent delegative trust in users who might otherwise develop it. This is the transparency paradox formalized in Claim 3: transparency builds evaluative trust ($\partial E/\partial \tau > 0$) but erodes the conditions under which delegative trust can form ($\partial \delta/\partial \tau < 0$). Recent experimental evidence supports this prediction: Schilke and Reimann (2025) find across thirteen experiments that disclosing AI usage erodes trust even when performance is held constant, consistent with the claim that transparency activates evaluative assessment while foreclosing the delegative mode.

The CFPB company-level analysis provides suggestive naturalistic evidence for this mechanism. Among mortgage companies with 50 or more complaints, the proportion of delegative complaints ranged from 3.6% to 37.8% — a 34.2 percentage-point spread. While exploratory, this variation suggests that institutional behavior — not just product type — shapes which trust mode consumers activate. Companies whose servicing practices create conditions for experienced vulnerability (long hold times on modification decisions, opaque escrow management, inconsistent communication) may generate more delegative complaints precisely because they have first created and then violated the conditions for delegative trust. This is a hypothesis for future research, not a conclusion from the current data.

The practical implication for robo-advisory platforms is specific: if the goal is long-term client retention rather than initial adoption, platforms should invest in fiduciary framing, discretionary authority structures, and the institutional conditions that permit delegative trust to form over time. This means less emphasis on real-time portfolio dashboards and per-trade explanations, and more emphasis on the relationship-like features that signal the platform's acceptance of fiduciary responsibility. The platforms that have best retained assets under management may already be doing this intuitively; our framework provides the theoretical basis for what they are doing and why it works. Multiple independent research programs are converging on the conclusion that trust in AI is irreducibly multidimensional (Glikson and Woolley, 2020; Hermann and Alberhasky, 2026; Huang et al., 2026), but these contributions lack a philosophical foundation that explains *why* the dimensions separate and *what* each dimension's distinct reactive

attitudes, update rules, and behavioral signatures are — which is precisely what the evaluative–delegative framework provides.

6.5 Limitations

Several limitations qualify the conclusions drawn above.

First, the evaluative trust composite in the Prolific experiment demonstrated poor internal reliability ($\alpha = .429$, two items). This attenuates any effects of evaluative trust on dependent variables and means that the null findings for evaluative trust in the H3 and H4 regressions cannot be interpreted as strong evidence of absence. The evaluative dimension may predict algorithm aversion and exit intent in ways our measurement was unable to detect. Future research should develop multi-item evaluative trust scales for algorithmic advisory contexts.

Second, the human validation of the CFPB classification involved a single coder (the author) rating 100 narratives. While categorical agreement with the AI models was fair ($\kappa = 0.264$–0.361) and improved dramatically at high confidence (93.3% agreement), a single coder cannot establish inter-rater reliability in the conventional sense. The primary disagreement pattern — the human coding more narratives as unclassifiable — is systematic and interpretable, but future work should employ multiple independent human coders to better calibrate the AI classification system.

Third, the CFPB sample is censored toward consumers who have already experienced relationship breakdowns severe enough to motivate a formal complaint. The negative correlation between delegative and evaluative scores ($r = -.59$) likely reflects this censoring rather than a true inverse relationship between the constructs, as argued in the results section (Section 4). But the censored sample also means we observe trust modes primarily in their violated state, which limits what the data can tell us about trust formation and maintenance.

Fourth, the robo-advisor experience predictor — the strongest evidence for Hardin's experiential-learning mechanism — is correlational. We cannot rule out that pre-existing individual differences (risk tolerance, technology comfort, financial sophistication) drive both the decision to use robo-advisors and the tendency to report higher delegative trust, rather than the experience itself building delegative trust over time. Longitudinal designs that track trust formation through early interactions with algorithmic advisors would provide stronger evidence.

Fifth, the manipulation failure means that causal claims about the effect of institutional framing on trust mode are not supported by the experimental data. The individual-differences analyses (H3, H4) are associational: they show that delegative trust *predicts* behavioral outcomes, not that it *causes* them. The theoretical framework provides the causal architecture, and the CFPB naturalistic data provide ecological validity, but the experimental causal evidence that a fully pre-registered design was intended to provide is absent.

Finally, the CFPB database does not cover investment advisory services (SEC/FINRA jurisdiction), and the Prolific experiment uses hypothetical vignettes rather than real investment decisions with financial consequences. Neither study observes actual investor behavior in response to actual portfolio losses. The triangulation across methods partially addresses this concern — naturalistic text and controlled experiment, banking products and algorithmic investment advice — but the ultimate test of the framework requires observation of real advisory relationships through adverse market conditions.

7 Conclusion

This paper introduced a two-dimensional framework for understanding trust in financial advisory relationships, distinguishing evaluative trust (evidence-conditional performance assessment) from delegative trust (the grant of discretionary authority with acceptance of opacity). Three empirical contributions support the framework's core claims.

First, dimensional separability is demonstrated across methods. In naturalistic CFPB complaint narratives, delegative trust indicators track product characteristics (relationship opacity and duration) while evaluative indicators remain flat — a pattern inconsistent with unidimensional models. In the Prolific experiment, the two trust composites share only 6% of their variance and predict non-overlapping sets of behavioral outcomes.

Second, delegative trust predicts resilience to both exit and algorithm aversion after adverse performance, while evaluative trust predicts neither. The H3 result — that delegative trust is the dominant predictor of resistance to algorithm aversion — reverses the pre-registered prediction but provides stronger evidence for the framework: algorithm aversion is an evaluative phenomenon, and resistance to it comes from the delegative dimension.

Third, the framework resolves a decade-long apparent contradiction in the algorithmic decision-making literature. Algorithm appreciation (Logg et al., 2019), algorithm aversion (Dietvorst et al., 2015), and the absence of betrayal aversion toward algorithms (Kormylo et al., 2026) are each manifestations of the evaluative dimension operating under different signal conditions. The second dimension — delegative trust — has been absent from the experimental literature because the standard paradigms do not create conditions for its formation.

The concept of delegative foreclosure, derived from the intersection of our framework with Hardin's (1993) street-level epistemology, identifies a mechanism the existing literature cannot name: the structural closure of a trust mode toward a category of trustee, distinct from the downward revision of a trust estimate. This concept explains both the persistence of algorithm aversion in the face of superior performance evidence and the failure of our own manipulation to shift trust modes through framing. Trust postures toward algorithmic advisors are learned through experience, not installed by institutional language. The robo-advisor experience predictor in the Prolific data ($B = 0.729$, $p = .001$) provides correlational support for this experiential-learning mechanism.

For practitioners, the framework inverts a common design assumption. The robo-advisory industry's emphasis on transparency, explainability, and performance dashboards is optimized for evaluative trust — and may actively undermine the delegative trust that predicts long-term client retention. Platforms seeking to reduce algorithm aversion and exit should invest not in better performance communication but in the institutional conditions that permit delegative trust to form: fiduciary framing, discretionary authority structures, and sustained interaction that allows vulnerability to be experienced as safe.

For regulators, the transparency paradox formalized in Claim 3 poses a genuine dilemma. Explainability mandates that make algorithmic decisions more legible may shift users from delegative to evaluative trust equilibria, increasing sensitivity to short-term performance fluctuations and raising the probability of exit — the opposite of the regulatory intent. This does not argue against transparency as a value, but it suggests that disclosure regimes should be designed with attention to their effects on trust mode, not merely on information availability.

The paper's limitations — particularly the evaluative trust measurement problem and the manipula-

tion failure — constrain the strength of causal claims. But the convergence of naturalistic and experimental evidence, across different populations, products, and measurement approaches, establishes the evaluative/delegative distinction as a productive framework for understanding trust in financial advisory relationships. The faithful machine is not the one that performs best on benchmarks. It is the one that earns, and honors, the grant of discretionary authority.

References

Baier, A. (1986). Trust and antitrust. *Ethics*, 96(2):231–260.

Belanche, D., Casaló, L. V., and Flavián, C. (2019). Artificial intelligence in FinTech: Understanding robo-advisors adoption among customers. *Industrial Management & Data Systems*, 119(7):1411–1430.

Berg, J., Dickhaut, J., and McCabe, K. (1995). Trust, reciprocity, and social history. *Games and Economic Behavior*, 10(1):122–142.

Bigman, Y. E. and Gray, K. (2018). People are averse to machines making moral decisions. *Cognition*, 181:21–34.

Burton, J. W., Stein, M.-K., and Jensen, T. B. (2020). A systematic review of algorithm aversion in augmented decision making. *Journal of Behavioral Decision Making*, 33(2):220–239.

Carter, J. A., Spencer, E., and Simion, M. (2020). Empirical and philosophical reflections on trust. In *Trust in Epistemology*. Oxford University Press.

Castelo, N., Bos, M. W., and Lehmann, D. R. (2019). Task-dependent algorithm aversion. *Journal of Marketing Research*, 56(5):809–825.

Chacon, A., Kausel, E. E., and Reyes, T. (2022). A longitudinal approach for understanding algorithm use. *Journal of Behavioral Decision Making*, 35(4):e2275.

Dietvorst, B. J., Simmons, J. P., and Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1):114–126.

Dietvorst, B. J., Simmons, J. P., and Massey, C. (2018). Overcoming algorithm aversion: People will use imperfect algorithms if they can (even slightly) modify them. *Management Science*, 64(3):1155–1170.

Foerster, S., Linnainmaa, J. T., Melzer, B. T., and Previtero, A. (2017). Retail financial advice: Does one size fit all? *Journal of Finance*, 72(4):1441–1482.

Gennaioli, N., Shleifer, A., and Vishny, R. (2015). Money doctors. *Journal of Finance*, 70(1):91–114.

Glikson, E. and Woolley, A. W. (2020). Human trust in artificial intelligence: Review of empirical research. *Academy of Management Annals*, 14(2):627–660.

Grossman, S. J. and Hart, O. D. (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. *Journal of Political Economy*, 94(4):691–719.

Guiso, L., Sapienza, P., and Zingales, L. (2004). The role of social capital in financial development. *American Economic Review*, 94(3):526–556.

Guiso, L., Sapienza, P., and Zingales, L. (2008). Trusting the stock market. *Journal of Finance*, 63(6):2557–2600.

Gurun, U. G., Stoffman, N., and Yonker, S. E. (2018). Trust busting: The effect of fraud on investor behavior. *Review of Financial Studies*, 31(4):1341–1376.

Hardin, R. (1993). The street-level epistemology of trust. *Politics & Society*, 21(4):505–529.

Hardin, R. (2002). *Trust and Trustworthiness*. Russell Sage Foundation, New York.

Hawley, K. (2014). Trust, distrust and commitment. *Noûs*, 48(1):1–20.

Hermann, E. and Alberhasky, M. (2026). The trusted partner for financial decision making: Romantic partner or AI? *Computers in Human Behavior*, 177:108877.

Holton, R. (1994). Deciding to trust, coming to believe. *Australasian Journal of Philosophy*, 72(1):63–76.

Hopp, F. R., Fisher, J. T., Cornell, D., Huskey, R., and Weber, R. (2021). The extended moral foundations dictionary (eMFD): Development and applications of a crowd-sourced approach to extracting moral intuitions from text. *Behavior Research Methods*, 53:232–246.

Huang, S., Chang, C.-Y., Park, S., Dinh, M. C., and Zhu, J. N. Y. (2026). Advice by algorithms: Dual dimensions of trust in AI financial service adoption. *Journal of Retailing and Consumer Services*, 88:104509.

Jensen, M. C. and Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4):305–360.

Johnson, N. D. and Mislin, A. A. (2011). Trust games: A meta-analysis. *Journal of Economic Psychology*, 32(5):865–889.

Jussupow, E., Benbasat, I., and Heinzl, A. (2024). An integrative perspective on algorithm aversion and appreciation in decision-making. *MIS Quarterly*, 48(4):1575–1612.

Kormylo, C., Adjerid, I., Ball, S. B., and Dogan, C. (2026). Till tech do us part: Betrayal aversion and its role in algorithm use. *Management Science*, 72(1):343–367.

Landis, J. R. and Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.

Lee, J. D. and See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1):50–80.

Lee, K., Cho, W., Woo, H.-G., and de Jong, S. (2026). When do individuals believe in themselves rather than in artificial intelligence? Insights from longitudinal investigations in corporate credit-rating contexts. *Journal of Management Studies*, 63(2):399–437.

Logg, J. M., Minson, J. A., and Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151:90–103.

Longoni, C., Bonezzi, A., and Morewedge, C. K. (2019). Resistance to medical artificial intelligence. *Journal of Consumer Research*, 46(4):629–650.

Lusardi, A. and Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and evidence. *Journal of Economic Literature*, 52(1):5–44.

Madhavan, P. and Wiegmann, D. A. (2007). Similarities and differences between human–human and human–automation trust: An integrative review. *Theoretical Issues in Ergonomics Science*, 8(4):277–301.

Mahmud, H., Islam, A. K. M. N., Ahmed, S. I., and Smolander, K. (2022). What influences algorithmic decision-making? A systematic literature review on algorithm aversion. *Technological Forecasting and Social Change*, 175:121390.

Mayer, R. C., Davis, J. H., and Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3):709–734.

McAllister, D. J. (1995). Affect- and cognition-based trust as foundations for interpersonal cooperation in organizations. *Academy of Management Journal*, 38(1):24–59.

Muir, B. M. (1994). Trust in automation: Part I. Theoretical issues in the study of trust and human intervention in automated systems. *Ergonomics*, 37(11):1905–1922.

Nickel, P. J. (2012). Trust in technological systems. In *The Philosophy of Trust*. Oxford University Press.

O'Neill, O. (2002). *A Question of Trust: The BBC Reith Lectures 2002*. Cambridge University Press, Cambridge.

O'Neill, O. (2014). Trust, trustworthiness, and accountability. In Morris, N. and Vines, D., editors, *Capital Failure: Rebuilding Trust in Financial Services*. Oxford University Press, Oxford.

Pettit, P. (1995). The cunning of trust. *Philosophy & Public Affairs*, 24(3):202–225.

Rossi, A. G. and Utkus, S. P. (2020). Who benefits from robo-advising? Evidence from machine learning. *Working Paper*.

Sapienza, P. and Zingales, L. (2012). A trust crisis. *International Review of Finance*, 12(2):123–131.

Schilke, O. and Reimann, M. (2025). The transparency dilemma: How AI disclosure erodes trust. *Organizational Behavior and Human Decision Processes*, 188:104405.

Strawson, P. F. (1962). Freedom and resentment. *Proceedings of the British Academy*, 48:1–25.

A CFPB Classification Prompt

The following prompt was used identically for both Claude 3.5 Sonnet (Anthropic) and GPT-4o (OpenAI) classifications. Each model received the prompt once per narrative, with temperature set to 0.0 (deterministic output) and a maximum output length of 300 tokens. No product type or company name was passed to the classifier, preventing hypothesis leakage. The prompt consists of a system prompt defining the construct framework and a user prompt specifying the required output format.

A.1 System Prompt

```
You are a research assistant classifying consumer financial
complaints for an academic study on trust in financial
relationships.

The study distinguishes two modes of trust failure:

EVALUATIVE TRUST FAILURE: The consumer's complaint is primarily
about unmet performance expectations, functional problems, or
verification failures. The consumer treats the financial
institution as a service provider that failed to deliver. Language
focuses on: incorrect charges, broken processes, unmet promises,
poor service quality, failure to perform as advertised. The
emotional register is frustration, dissatisfaction, or anger at
incompetence. The consumer's implicit stance is: "You didn't do
what you were supposed to do."

DELEGATIVE TRUST FAILURE: The consumer's complaint reflects
betrayal of a relationship in which they had granted the
institution discretionary authority or relied on it as a fiduciary.
Language focuses on: betrayal, abandonment, broken loyalty, abuse
of the relationship, violation of good faith, taking advantage of
vulnerability. The emotional register is moral outrage, a sense of
personal violation, or grief at a relationship destroyed. The
consumer's implicit stance is: "I trusted you and you betrayed me."

UNCLASSIFIABLE: The complaint does not clearly reflect either trust
mode. This includes: template/form letters, pure factual disputes
with no emotional content, identity theft reports where the
consumer has no relationship with the institution, complaints too
brief or unclear to assess, or complaints that mix both modes
without a dominant one.

Important: Many complaints will be unclassifiable. That is
expected. Do not force a classification. Only assign EVALUATIVE or
DELEGATIVE when the language clearly reflects one mode.
```

A.2 User Prompt Template

For each narrative, the following user prompt was appended after the system prompt, with the complaint text inserted in place of the placeholder:

```
Classify this consumer financial complaint. Respond in exactly
this JSON format:

{
  "classification": "EVALUATIVE" | "DELEGATIVE" | "UNCLASSIFIABLE",
  "confidence": <float 0.0 to 1.0>,
  "primary_evidence": "<one sentence quoting or paraphrasing the
    specific language that drove your classification>",
  "secondary_signals": "<one sentence noting any additional
    signals, or 'none'>",
  "delegative_score": <float 0.0 to 1.0>,
  "evaluative_score": <float 0.0 to 1.0>
}

The delegative_score and evaluative_score are independent
dimensions, not a zero-sum scale. A complaint can score high on
both (mixed), low on both (unclassifiable), or high on one and
low on the other (clean classification).

COMPLAINT:
"""
[Narrative text inserted here]
"""
```

A.3 Classification Parameters

Both models were configured with identical parameters to ensure comparability. Table A.1 summarizes the classification pipeline settings.

Table A.1: Classification pipeline parameters.

Parameter	Claude 3.5 Sonnet	GPT-4o
Model version	`claude-3-5-sonnet-20241022`	`gpt-4o`
Temperature	0.0	0.0
Max output tokens	300	300
Concurrent requests	20	5
Retry logic	Exponential backoff (5 attempts)	Exponential backoff (5 attempts)
Narratives classified	8,943	8,943

B Prolific Survey Instrument

The following instrument was administered via Qualtrics to participants recruited through the Prolific platform. Demographics and financial literacy were administered before condition assignment. Items are reproduced exactly as presented to participants, with correct answers indicated in footnotes.

B.1 Part 1: Demographics and Financial Literacy

B.1.1 Financial Literacy (Lusardi and Mitchell, 2014)

FL1. Suppose you had $100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?

1. More than $102[3]
2. Exactly $102
3. Less than $102
4. Don't know

FL2. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

1. More than today
2. Exactly the same
3. Less than today[4]
4. Don't know

FL3. Please tell me whether this statement is true or false: "Buying a single company's stock usually provides a safer return than a stock mutual fund."

1. True
2. False[5]
3. Don't know

Scoring: Sum of correct responses (0–3).

[3] Correct answer for FL1.
[4] Correct answer for FL2.
[5] Correct answer for FL3.

B.1.2 Background Information

D1. What is your age? _____ years

D2. What is your gender?

1. Male
2. Female
3. Non-binary
4. Prefer not to say

D3. What is your annual household income before taxes?

1. Under $25,000
2. $25,000–$49,999
3. $50,000–$74,999
4. $75,000–$99,999
5. $100,000–$149,999
6. $150,000–$199,999
7. $200,000 or more

D4. What is your highest level of education?

1. Less than high school
2. High school diploma or GED
3. Some college
4. Bachelor's degree
5. Graduate degree

D5. Which of the following investment experiences have you had? (Check all that apply)

☐ Used a robo-advisor (e.g., Betterment, Wealthfront)

☐ Worked with a human financial advisor

☐ Self-directed investing through a brokerage account

☐ Employer retirement plan (401k, 403b)

☐ No investment experience

D6. [Attention Check] How much do you agree that attention to detail is important in financial decisions?
(This question is checking if you are reading carefully. Please select "Somewhat disagree" for this item.)

1. Strongly Disagree
2. Moderately Disagree
3. Somewhat Disagree[6]
4. Neither Agree nor Disagree
5. Somewhat Agree
6. Moderately Agree
7. Strongly Agree

B.2 Part 2: Condition Vignettes

Participants were randomly assigned to one of three conditions. All vignettes describe the same technological agent (WealthPath) with matched structural features (0.25% annual fee, $1,000 minimum, SIPC protection, custodial banking). Only the trust framing varies.

B.2.1 Condition A — Relationship/Fiduciary

> "WealthPath is your dedicated digital investment partner, designed to serve as a steward for your long-term financial goals. The platform's algorithms are built to act in your best interest, with full discretionary authority to adjust your portfolio as market conditions change. WealthPath is committed to building a lasting relationship with you through clear communication about its investment approach and personalized service aligned with your financial future. The platform is designed to guide you through market volatility with steady, disciplined investing on your behalf. WealthPath charges a 0.25% annual management fee and requires a $1,000 minimum investment. Account assets are held at a major custodian bank with SIPC protection up to $500,000."

B.2.2 Condition B — Performance/Analytics

> "WealthPath is a digital investment platform that uses systematic strategies to manage your portfolio efficiently. Our algorithms continuously analyze market data and economic indicators to make disciplined investment decisions based on quantitative analysis rather than emotion. WealthPath follows consistent rebalancing schedules and maintains proper diversification across different types of investments. The platform provides detailed reports showing exactly how your money is allocated and how your investments are performing compared to market benchmarks. You can track your portfolio's progress with clear analytics that show which decisions are working and why. WealthPath charges a 0.25% annual management fee and requires a $1,000 minimum investment. Account assets are held at a major custodian bank with SIPC protection up to $500,000."

[6] Instructed response for attention check.

B.2.3 Condition C — Neutral Control

> "WealthPath is a digital investment platform that automatically manages your portfolio using algorithmic trading strategies. The platform creates diversified portfolios across stocks, bonds, and ETFs based on your risk tolerance and investment timeline. WealthPath charges a 0.25% annual management fee and requires a $1,000 minimum investment. The platform rebalances your portfolio quarterly and provides monthly performance reports through their mobile app. WealthPath has been operating since 2018 and currently manages portfolios for individual investors across all 50 states. Account assets are held at a major custodian bank with SIPC protection up to $500,000."

B.3 Part 3: Stage 1 Items (Evaluative and Delegative Trust)

Administered immediately after vignette. All items use a 7-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree).

Evaluative Trust ($\alpha = .429$):

1. "I would want to regularly check WealthPath's performance against market benchmarks."
2. "I would trust WealthPath primarily because I can verify how it's performing."

Delegative Trust ($\alpha = .807$):

3. "I would be comfortable giving WealthPath full discretion to adjust my portfolio as it sees fit."
4. "I would be willing to let WealthPath make investment decisions without needing to understand each one."
5. "Even if I could review every decision WealthPath makes, I probably wouldn't bother."

B.4 Part 4: Adverse Event

Identical across all conditions.

> "Three months after opening your WealthPath account, you receive this notification:
>
> *'Your portfolio declined 8.7% this quarter during a broad market downturn. The S&P 500 declined 8.0% over the same period. Our models have adjusted positions to reflect current market conditions. No action is required on your part.'*
>
> Your account balance shows a loss of $870 on your $10,000 investment."

B.5 Part 5: Comprehension Check

"Based on the notification you just read, how did WealthPath's performance compare to the overall stock market (S&P 500) during this period?"

1. WealthPath performed better than the market

2. WealthPath performed about the same as the market
3. WealthPath performed worse than the market[7]
4. I'm not sure

B.6 Part 6: Stage 2 Items (Reactive Attitudes and Behavioral Response)

Administered after comprehension check. All items use the same 7-point Likert scale.

Reactive Attitudes:

6a. "After receiving WealthPath's notification about the decline, I feel a sense of betrayal."

6b. "After receiving WealthPath's notification about the decline, I feel disappointed by the results."

Betrayal premium = Item 6a minus Item 6b.

Behavioral Response:

7a. "My first reaction is to research alternative investment platforms rather than wait for WealthPath's strategy to recover."

7b. "After this experience, I would prefer to manage my own investments or work with a human financial advisor rather than continue using an algorithm."

B.7 Scale Reference

Table B.1: 7-point Likert scale used for all attitude and trust items.

Value	Label
1	Strongly Disagree
2	Moderately Disagree
3	Somewhat Disagree
4	Neither Agree nor Disagree
5	Somewhat Agree
6	Moderately Agree
7	Strongly Agree

C Inter-Rater Reliability Details

This appendix reports the full inter-rater reliability analysis for the CFPB classification system. Three raters classified a confidence-stratified subsample of 100 narratives: one human coder (the author) and two large language models (Claude 3.5 Sonnet and GPT-4o). Full-sample concordance statistics are reported for the dual-model classification of all 8,943 narratives classified by both models.

[7] Correct answer for comprehension check.

C.1 Human Validation Sample

The 100-narrative subsample was stratified by Sonnet confidence level to ensure representation across the classification difficulty spectrum: 30 low-confidence narratives (Sonnet confidence < 0.75), 40 medium-confidence narratives (0.75–0.85), and 30 high-confidence narratives (> 0.85). The human coder classified each narrative independently using the same construct definitions provided to the AI models (Appendix A), recording a categorical classification (evaluative, delegative, or unclassifiable), continuous delegative and evaluative scores (0–1, using a slider input rounded to 0.1 increments), and a categorical confidence rating (high, medium, or low).

C.2 Pairwise Categorical Agreement

Table C.1: Pairwise categorical agreement on 100-narrative validation subsample.

Pair	Cohen's κ	95% CI	% Agreement	Interpretation
Human vs. Sonnet	0.264	[0.136, 0.397]	54.0%	Fair
Human vs. GPT-4o	0.361	[0.207, 0.507]	62.0%	Fair
Sonnet vs. GPT-4o	0.590	[0.442, 0.725]	77.0%	Moderate

Interpretation benchmarks follow Landis and Koch (1977): < 0.20 poor, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, 0.81–1.00 almost perfect.

C.3 Pairwise Continuous Score Correlations

Human scores were recorded using 0.1-increment slider input; LLM scores are continuous floats. Correlations are computed on raw values.

Table C.2: Pairwise correlations for delegative scores ($N = 100$).

Pair	Pearson r	p	Spearman ρ	p
Human vs. Sonnet	.488	2.65×10^{-7}	.377	1.11×10^{-4}
Human vs. GPT-4o	.511	5.69×10^{-8}	.372	1.40×10^{-4}
Sonnet vs. GPT-4o	.776	2.50×10^{-21}	.827	3.00×10^{-26}

Table C.3: Pairwise correlations for evaluative scores ($N = 100$).

Pair	Pearson r	p	Spearman ρ	p
Human vs. Sonnet	.308	1.83×10^{-3}	.400	3.68×10^{-5}
Human vs. GPT-4o	.402	3.31×10^{-5}	.458	1.62×10^{-6}
Sonnet vs. GPT-4o	.779	1.33×10^{-21}	.818	3.06×10^{-25}

C.4 Confusion Matrices

Table C.4: Confusion matrix: Human vs. Sonnet ($N = 100$).

	Sonnet			
Human	Evaluative	Delegative	Unclassifiable	Total
Evaluative	29	10	0	39
Delegative	1	4	0	5
Unclassifiable	34	1	21	56
Total	64	15	21	100

Table C.5: Confusion matrix: Human vs. GPT-4o ($N = 100$).

	GPT-4o			
Human	Evaluative	Delegative	Unclassifiable	Total
Evaluative	30	4	5	39
Delegative	1	4	0	5
Unclassifiable	24	4	28	56
Total	55	12	33	100

Table C.6: Confusion matrix: Sonnet vs. GPT-4o ($N = 100$).

	GPT-4o			
Sonnet	Evaluative	Delegative	Unclassifiable	Total
Evaluative	49	4	11	64
Delegative	5	8	2	15
Unclassifiable	1	0	20	21
Total	55	12	33	100

C.5 Confidence-Stratified Agreement

Table C.7: Pairwise agreement rates by Sonnet confidence stratum ($N = 100$).

Confidence Bin	N	Human–Sonnet	Human–GPT-4o	Sonnet–GPT-4o
Low (< 0.75)	30	30.0%	50.0%	50.0%
Medium (0.75–0.85)	40	42.5%	50.0%	82.5%
High (> 0.85)	30	93.3%	90.0%	96.7%

C.6 Classification Distribution Comparison

Table C.8: Classification distribution across raters ($N = 100$).

Category	Human	Sonnet	GPT-4o
Evaluative	39 (39%)	64 (64%)	55 (55%)
Delegative	5 (5%)	15 (15%)	12 (12%)
Unclassifiable	56 (56%)	21 (21%)	33 (33%)

The human coder classified substantially more narratives as unclassifiable (56%) than either AI model (Sonnet: 21%, GPT-4o: 33%), reflecting a higher threshold for inferring trust-relevant language from factual complaint descriptions.

C.7 Disagreement Pattern Analysis

Of the 100 narratives, 49 received unanimous agreement across all three raters. The remaining 51 disagreements followed systematic patterns (Table C.9).

Table C.9: Systematic disagreement patterns ($N = 51$ discordant cases).

Human	Sonnet	GPT-4o	Count
Unclassifiable	Evaluative	Evaluative	23
Unclassifiable	Evaluative	Unclassifiable	8
Evaluative	Delegative	Evaluative	4
Evaluative	Delegative	Delegative	4
Unclassifiable	Evaluative	Delegative	3
Evaluative	Evaluative	Unclassifiable	3
Evaluative	Delegative	Unclassifiable	2
Other patterns			4

The dominant disagreement pattern (23 of 51 cases, 45%) involves both AI models classifying a narrative as evaluative where the human coded it as unclassifiable. These cases typically involve factual complaints describing service failures (account holds, missing documents, disputed charges) without explicit evaluative language. The AI models inferred frustrated performance expectations from the complaint structure; the human coder required more explicit evaluative vocabulary to assign the classification.

C.8 Illustrative Disagreement Cases

Case 1 (Complaint #4121639, Checking/Savings, Chime Financial). A consumer reports that Chime placed a hold on their account containing unemployment back pay and that they sent all requested documentation. The human coded this as UNCLASSIFIABLE ("factual"), while both Sonnet and GPT-4o coded it as EVALUATIVE, interpreting the account hold despite compliance with documentation requests as an implicit performance failure. This illustrates the threshold

difference: the human required explicit evaluative language, while the models inferred it from the service failure structure.

Case 2 (Complaint #10717654, Mortgage, Mr. Cooper). A consumer reports missing loan documents, discrepancies in payoff quotes, and COVID deferral payments reported as delinquent. The human coded this as UNCLASSIFIABLE ("factual, no evaluative or delegative language"), while both models coded it as EVALUATIVE, interpreting the document failures and credit reporting errors as implicit performance complaints. Again, the complaint describes functional failures without using the language of frustrated expectations.

Case 3 (Complaint #7441427, Credit Card, Wells Fargo). A consumer reports disputed fraudulent charges that Wells Fargo repeatedly denied. The human coded this as EVALUATIVE (noting "frustrated expectations"), Sonnet coded it as DELEGATIVE (citing "I feel like I'm not being heard"), and GPT-4o coded it as EVALUATIVE (focusing on incorrect charges). This case illustrates genuine construct ambiguity: the complaint contains both evaluative elements (the fraud process failed) and delegative elements (the relationship feels unresponsive), and reasonable coders disagree about which mode dominates.

C.9 Full-Sample Dual-Model Concordance ($N = 8{,}943$)

For reference, the full-sample concordance statistics for the complete CFPB dataset are reported here and discussed in Section 4.

Table C.10: Full-sample concordance: Sonnet vs. GPT-4o ($N = 8{,}943$). Overall agreement: 7,369/8,943 (82.4%), $\kappa = 0.556$.

	GPT-4o			
Sonnet	Evaluative	Delegative	Unclassifiable	Total
Evaluative	6,037	351	575	6,963
Delegative	519	877	81	1,477
Unclassifiable	30	18	455	503
Total	6,586	1,246	1,111	8,943

Panel A: Mortgage (n = 2,974; agreement = 83.1%, κ = 0.571)

Table C.11: Per-product confusion matrices: Sonnet vs. GPT-4o (N = 8,943).

	GPT-4o			
Sonnet	Evaluative	Delegative	Unclassifiable	Total
Evaluative	1,996	170	108	2,274
Delegative	187	414	24	625
Unclassifiable	8	5	62	75
Total	2,191	589	194	2,974

Panel B: Checking/Savings (n = 2,985; agreement = 80.6%, κ = 0.530)

	GPT-4o			
Sonnet	Evaluative	Delegative	Unclassifiable	Total
Evaluative	1,958	106	264	2,328
Delegative	161	286	38	485
Unclassifiable	8	3	161	172
Total	2,127	395	463	2,985

Panel C: Credit Card (n = 2,984; agreement = 83.5%, κ = 0.560)

	GPT-4o			
Sonnet	Evaluative	Delegative	Unclassifiable	Total
Evaluative	2,083	75	203	2,361
Delegative	171	177	19	367
Unclassifiable	14	10	232	256
Total	2,268	262	454	2,984

Rows = Claude 3.5 Sonnet (`claude-3-5-sonnet-20241022`) classification. Columns = GPT-4o classification. Diagonal cells represent inter-model agreement. Credit card shows the highest agreement (83.5%) despite having the lowest delegative base rate, while checking/savings shows the lowest (80.6%), driven primarily by Sonnet–evaluative / GPT-4o–unclassifiable disagreements.

Continuous score correlations between models (N = 8,943): delegative r = .703 (ρ = .738), evaluative r = .680 (ρ = .713), all $p < .001$.

C.10 Full-Sample Confidence-Stratified Concordance

Table C.12: Confidence-stratified inter-model agreement and human validation ($N_{\text{full}} = 8{,}943$; $N_{\text{human}} = 100$).

		Full sample		Human validation		
Stratum	Range	N	Agreement	N	Human–Sonnet	Human–GPT-4o
Low	0.51–0.78	3,759	68.8%	30	30.0%	50.0%
Medium	0.78–0.83	3,124	87.7%	40	42.5%	50.0%
High	0.85–0.97	2,060	99.2%	30	93.3%	90.0%
Overall	—	8,943	82.4%	100	54.0%	62.0%

Full-sample strata are terciles of mean classifier confidence (average of Sonnet and GPT-4o confidence ratings). Human validation strata use Sonnet confidence bins from the stratified sampling design (low < 0.75, medium 0.75–0.85, high > 0.85). Strata boundaries differ slightly between the two analyses because the full sample uses distributional terciles while the validation sample used pre-specified thresholds. Full-sample Cohen's κ: low = 0.356, medium = 0.652, high = 0.970, overall = 0.556. At high confidence, both inter-model and human–model agreement converge above 90%, supporting reliance on high-confidence automated classifications.

D Supplementary Tables

Tables D.1–D.5 supplement the CFPB complaint analysis (Section 4). Tables D.6–D.10 supplement the Prolific experiment (Section 5). Significance levels: * $p < .05$, ** $p < .01$, *** $p < .001$.

D.1 CFPB Supplementary Tables

Table D.1: eMFD moral foundations validation: evaluative vs. delegative classifications ($N_{\text{eval}} = 6{,}963$; $N_{\text{deleg}} = 1{,}477$).

Foundation	Evaluative M (SD)	Delegative M (SD)	Cohen's d	t	p
Care	.02673 (.00598)	.02807 (.00537)	0.228	8.52	$< .001$***
Fairness	.03053 (.00696)	.03156 (.00606)	0.151	5.78	$< .001$***
Loyalty	.02311 (.00489)	.02434 (.00434)	0.256	9.65	$< .001$***
Authority	.02429 (.00549)	.02478 (.00463)	0.091	3.55	$< .001$***
Sanctity	.01999 (.00461)	.02115 (.00423)	0.255	9.42	$< .001$***

Values are mean eMFD probability scores (per-word moral foundation density) with standard deviations in parentheses. Classifications based on Sonnet model. Delegative complaints score significantly higher on all five moral foundations, with the strongest effects for loyalty ($d = 0.256$) and sanctity ($d = 0.255$). Welch's t-test (unequal variances); all comparisons survive Bonferroni correction ($\alpha/5 = .01$). Cohen's d computed with pooled standard deviation.

Table D.2: eMFD scores by product type ($N = 8{,}991$). Values are mean probability scores with standard deviations in parentheses.

Foundation	Mortgage	Checking/Savings	Credit Card
Care	0.02653 (0.00597)	0.02720 (0.00618)	0.02742 (0.00601)
Fairness	0.03098 (0.00692)	0.03088 (0.00716)	0.03055 (0.00688)
Loyalty	0.02320 (0.00469)	0.02396 (0.00521)	0.02296 (0.00493)
Authority	0.02364 (0.00496)	0.02463 (0.00578)	0.02512 (0.00559)
Sanctity	0.01995 (0.00449)	0.02046 (0.00495)	0.02044 (0.00481)

Table D.3: Narrative length (characters) by trust classification and product type.

Panel A: Sonnet classifications

Product	Classification	*N*	Mean	Median	SD
Mortgage	Evaluative	2,274	1,590	1,122	1,751
	Delegative	625	2,144	1,589	2,161
	Unclassifiable	75	1,637	715	2,821
Checking/Savings	Evaluative	2,328	1,230	905	1,089
	Delegative	485	1,889	1,419	1,905
	Unclassifiable	172	664	456	604
Credit Card	Evaluative	2,361	1,305	963	1,210
	Delegative	367	1,798	1,509	1,129
	Unclassifiable	256	867	470	1,103

Panel B: GPT-4o classifications

Product	Classification	*N*	Mean	Median	SD
Mortgage	Evaluative	2,191	1,539	1,129	1,435
	Delegative	589	2,562	1,855	3,007
	Unclassifiable	194	1,022	692	1,050
Checking/Savings	Evaluative	2,127	1,264	955	1,112
	Delegative	395	2,047	1,459	2,034
	Unclassifiable	463	857	620	745
Credit Card	Evaluative	2,268	1,305	1,002	1,124
	Delegative	262	2,236	1,787	1,745
	Unclassifiable	454	924	612	960

Panel A: Regression coefficients

Table D.4: OLS regression: narrative length as predictor of Sonnet delegative score ($N = 8{,}943$).

Predictor	B	SE	t	p	95% CI	
Intercept	−0.243	0.021	−11.40	< .001***	−0.285	−0.201
log(Character count)	0.079	0.003	25.90	< .001***	0.073	0.085
Credit card (vs. Checking/Savings)	−0.038	0.006	−6.65	< .001***	−0.049	−0.027
Mortgage (vs. Checking/Savings)	0.016	0.006	2.71	.007**	0.004	0.027

Panel B: Cross-product delegative gradient, raw vs. length-controlled

Product	Raw Mean Deleg. Score	Residualized Mean
Mortgage	0.334	+0.023
Checking/Savings	0.299	+0.008
Credit Card	0.263	−0.030
Gradient (max − min)	0.071	0.053

Gradient retained after length control: 74.3%. The cross-product delegative gradient survives length control; narrative length does not explain away the between-product differences in delegative trust.

Table D.5: Company-level delegative proportion for mortgage complaints (companies with ≥ 50 narratives, $N = 15$).

Company	Total	Eval.	Deleg.	Unclass.	Deleg. %
Wells Fargo & Co.	241	136	91	14	37.8
Bank of America	116	72	39	5	33.6
JPMorgan Chase & Co.	95	66	27	2	28.4
Ocwen Financial	173	124	44	5	25.4
Nationstar Mortgage	81	60	20	1	24.7
U.S. Bancorp	56	40	13	3	23.2
Select Portfolio Servicing	107	81	23	3	21.5
Ditech Financial	71	56	15	0	21.1
Rocket Mortgage	63	49	12	2	19.0
Mr. Cooper Group	152	121	26	5	17.1
Specialized Loan Servicing	72	59	12	1	16.7
Freedom Mortgage	99	81	16	2	16.2
Shellpoint Partners	185	154	29	2	15.7
PennyMac Loan Services	52	44	7	1	13.5
LoanCare	83	80	3	0	3.6

Sorted by delegative proportion (descending). Delegative proportion ranges from 3.6% (LoanCare) to 37.8% (Wells Fargo), a spread of 34.2 percentage points. Classifications based on Sonnet model; concordant-only analysis yields similar ordering.

D.2 Prolific Supplementary Tables

Table D.6: Robustness check: hypothesis tests under strict attention exclusion ($N = 342$).

Hypothesis	Prediction	Key Statistic	p	Supported
H1a (Eval: B $\geq$ A)	Higher evaluative in B	B $\geq$ A	.434	No
H1b (Deleg: A $\geq$ B)	Higher delegative in A	A $\geq$ B	.147	No
H2 (Betrayal premium)	Higher in A than B	A $>$ B	.483	No
H3 (Algorithm aversion)	Eval predicts, Deleg does not	$\beta_{\text{eval}} = -.058$	.462	No
		$\beta_{\text{deleg}} = -.322$		
H4 (Deleg. resilience)	Deleg predicts lower exit	$\beta_{\text{deleg}} = -.207$	.001**	Yes
H5 (Literacy $\times$ Cond.)	Interaction on delegative	$\beta_{\text{int}} = .082$	.854	No

Excludes participants who failed the attention check (D6). All results match the direction and significance pattern of the primary analysis ($N = 354$).

Table D.7: Robustness check: hypothesis tests under strict comprehension exclusion ($N = 289$).

Hypothesis	Prediction	Key Statistic	p	Supported
H1a (Eval: B $\geq$ A)	Higher evaluative in B	B $\geq$ A	.389	No
H1b (Deleg: A $\geq$ B)	Higher delegative in A	A $\geq$ B	.157	No
H2 (Betrayal premium)	Higher in A than B	A $>$ B	.511	No
H3 (Algorithm aversion)	Eval predicts, Deleg does not	$\beta_{\text{eval}} = -.047$	.574	No
		$\beta_{\text{deleg}} = -.316$		
H4 (Deleg. resilience)	Deleg predicts lower exit	$\beta_{\text{deleg}} = -.161$	.021*	Yes
H5 (Literacy $\times$ Cond.)	Interaction on delegative	$\beta_{\text{int}} = -.269$	.606	No

Excludes participants who failed the comprehension check. All results match the direction and significance pattern of the primary analysis ($N = 354$).

Table D.8: Full OLS regression output for H3 (algorithm aversion) and H4 (general exit intent).

Predictor	Algorithm Aversion (H3) *B* (SE)	General Exit (H4) *B* (SE)
Intercept	5.573*** (0.671)	4.709*** (0.762)
Evaluative trust	−0.039 (0.077)	0.037 (0.087)
Delegative trust	−0.326*** (0.056)	−0.206** (0.064)
Condition A	0.073 (0.192)	−0.056 (0.218)
Condition B	−0.250 (0.186)	−0.132 (0.211)
Age	−0.006 (0.006)	0.005 (0.007)
Financial literacy	−0.191 (0.118)	−0.251 (0.134)
Robo-advisor exp.	−0.215 (0.234)	0.242 (0.266)
Comprehension (pass)	0.584** (0.204)	0.616** (0.232)
Attention check (pass)	1.179** (0.421)	0.577 (0.478)
R^2	0.151	0.054
Adj. R^2	0.129	0.029
N	354	354

Reference category for condition dummies: Condition C (neutral control). Standardized β for delegative trust: −0.304 (H3), −0.179 (H4). Standardized β for comprehension check: 0.149 (H3), 0.146 (H4).

Table D.9: Full OLS regression output for H5: financial literacy × condition interaction on delegative trust.

Predictor	*B*	SE	*t*	*p*
Intercept	3.435	0.577	5.96	< .001***
Condition A	−0.042	0.377	−0.11	.911
Condition B	0.259	0.368	0.70	.482
Financial literacy (high)	−0.407	0.318	−1.28	.201
Condition A × Fin. lit. (high)	0.072	0.433	0.17	.869
Condition B × Fin. lit. (high)	−0.109	0.424	−0.26	.797
Age	0.004	0.006	0.70	.484
Robo-advisor experience	0.729	0.227	3.21	.001**
Comprehension check (pass)	0.066	0.201	0.33	.743
Attention check (pass)	0.418	0.415	1.01	.314
R^2	0.052			
Adj. R^2	0.027			
N	354			

Reference category: Condition C, low financial literacy (score 0–1). High financial literacy defined as score 2–3 on the Lusardi–Mitchell Big Three. The condition × literacy interactions are non-significant, indicating that the framing effects (or lack thereof) do not vary by financial sophistication.

Panel A: E1 — Variance prediction (Levene's test)

Table D.10: Exploratory analyses E1–E3 ($N = 354$).

DV	$F(2, 351)$	p	σ^2_A / σ^2_B / σ^2_C	Supported
Delegative trust	0.318	.728	1.94 / 2.18 / 1.92	No

Panel B: E2 — Comprehension check as signal of trust mode

DV	Test	Pass M	Fail M	p	Direction
Delegative trust	t-test	3.90	3.97	.739	Fail > Pass
Evaluative trust	t-test	5.34	5.11	.110	Pass > Fail

Panel C: E3 — Robo-advisor experience × condition interactions

	Cond. A × Robo		Cond. B × Robo	
DV	B	p	B	p
Evaluative trust	0.358	.408	0.164	.680
Delegative trust	0.211	.720	−0.072	.894
Betrayal	0.728	.284	−0.667	.285
Disappointment	1.174	.022*	0.645	.168
Betrayal premium	−0.446	.510	−1.313	.035*
General exit	0.598	.383	0.584	.354
Algorithm aversion	0.535	.391	−0.519	.365

E1: Prediction that Condition A produces higher variance on delegative trust was not supported. *E2:* Comprehension check pass rates were comparable across conditions (A: 80.9%, B: 81.1%, C: 83.0%; $N_{\text{pass}} = 289$, $N_{\text{fail}} = 65$). Directional trends are consistent with E2 predictions but do not reach significance. *E3:* Two of 14 interaction tests reach significance: robo-advisor experience moderates disappointment in Condition A ($p = .022$) and betrayal premium in Condition B ($p = .035$). Given 14 tests, these may reflect chance.